

GREENOPS
IPS12 – INTELLIGENT DATA CENTER RESOURCE
OPTIMIZATION AND SUSTAINABILITY MANAGEMENT
PLATFORM

PHASE I PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING



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OCTOBER 2026

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BONAFIDE CERTIFICATE

Certified that this project report titled “**IPS12 – INTELLIGENT DATA CENTER RESOURCE OPTIMIZATION AND SUSTAINABILITY MANAGEMENT PLATFORM**” is the bonafide work of AISHWARYA A (230701015), AISHWARYA M (230701016), KAVIYA MADHIRAJU (230701148), and KARTHIKHA SRE M (230701143), who carried out the project work under my supervision. Certified further that, to the best of my knowledge the work herein does not form part of any thesis or dissertation on the basis of which degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal(s):

Goal No : 13 - Climate Action - Take urgent action to combat climate change and its impacts by reducing emissions and strengthening climate resilience

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ABSTRACT

Data centers consume vast electricity and water, with significant waste from idle servers, stale or duplicate storage, and unoptimized cooling. Existing sustainability tools often monitor resources in isolation and report metrics without simulating the impact of optimization actions before execution. GreenOps bridges this gap by ingesting multi-domain telemetry (compute, power, cooling, storage) to compute live metrics like PUE, WUE, energy, cost, and carbon. A statistically derived rule engine identifies idle servers and storage inefficiencies using server-type-specific thresholds. Four machine learning models support the platform: two validation models trained on 10,000 telemetry records estimate CPU utilization and power draw, while two forecasting models predict near-term workload demand. Before proposing server consolidation, GreenOps executes a what-if simulation evaluating post-move CPU, memory, network, and thermal headroom against live and forecasted limits. Recommendations are ranked by weighted sustainability and risk factors, logged for auditing, and restricted via role-based access control. By keeping final execution decisions under human operator control, GreenOps delivers transparent decision support for data center sustainability. This report details the architecture, modelling, recommendation logic, and experimental validation of the platform.

Keywords: GreenOps, Data Center Sustainability, Resource Optimization, Telemetry Analytics, Machine Learning, Workload Forecasting, What-If Simulation, PUE, WUE, Server Consolidation

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.)THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. MANORANJINI, Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work.

We convey our sincere and deepest gratitude to our internal guide, **Ms. R. AKILADEVI, M.E**, Assistant Professor(SS), Department of Computer Science and Engineering for her valuable guidance throughout the course of the project. We thank our Project Coordinator, **Mr. S. MADHUSUDHANAN, Ph.D.**, Associate Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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LIST OF ABBREVIATIONS

PUE	Power Usage Effectiveness
WUE	Water Usage Effectiveness
CPU	Central Processing Unit
kWh	Kilowatt-hour
kg	Kilogram
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination
ML	Machine Learning
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token
RBAC	Role-Based Access Control
ESG	Environmental, Social and Governance
NIC	Network Interface Card
SLA	Service Level Agreement
PDU	Power Distribution Unit
ORM	Object-Relational Mapping
CRUD	Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Data centers now account for a rapidly growing share of global electricity demand and industry studies consistently find that a significant fraction of that consumption is not doing useful work: servers sit idle at low utilization while still drawing near-full power, storage volumes accumulate stale and duplicated data that is never accessed again and cooling systems run without any real-time awareness of the load they are actually cooling. This waste is not exotic or hard to find; it is observable in ordinary infrastructure telemetry but it is rarely acted upon because the information needed to see it is scattered across separate monitoring systems for compute, power, cooling and storage and because most existing tools stop at reporting a dashboard number rather than proposing and safety-checking a concrete action.

GreenOps is a full-stack observability and recommendation platform built to address this gap directly. It structures every telemetry stream through the same five-stage pipeline: observing, analyzing, predicting, evaluating and recommending. The system never executes an infrastructure change on its own. It detects waste using transparent rule-based logic, validates current server behavior using machine learning models trained on real historical telemetry and forecasts near-term demand using models trained on the system's own accumulated history, simulates the safety and impact of a proposed consolidation before it happens and hands a ranked, itemized recommendation to a human operator who makes the final call. Every decision the operator takes is logged with the exact estimate that was shown at the time, producing an auditable trail suitable for sustainability and ESG reporting.

1.2 PROBLEM STATEMENT

Operators of data center infrastructure need a way to keep answering a few practical questions on an ongoing basis. Which servers are actually idle and worth consolidating? Which stored data is stale or duplicated and could be archived or deduplicated? Which storage volumes are over-provisioned compared to what they actually hold? And if a workload is moved onto a candidate server, is that move actually safe given the server's current and near future load, memory, network and thermal headroom? Is the data center sustainable? How much water is being used and how much carbon is being emitted? What to do to make it more sustainable?

Most existing tools don't answer all of this. They usually cover only one piece, compute or power or cooling or storage in isolation. They apply a single flat threshold no matter what type of hardware is involved and they don't simulate what would actually happen before recommending an action. Most tools don't tell us what's a directly measured value, what's a modelled estimate, and what simply isn't supported yet. That distinction matters a lot once the output is being used to justify a real infrastructure change or reported externally as a sustainability number.

1.3 OBJECTIVES

- To ingest and validate telemetry across four domains (server compute, power, cooling and storage) on a continuous, per-server basis.
- To compute live, formula-based sustainability metrics (PUE, WUE, energy, cost and carbon) and clearly separate genuinely measured ratios from labelled estimates.
- To detect operational waste, including idle servers, stale data, duplicated data and over-provisioned storage, using a rule engine with

per-server-type thresholds derived statistically from real historical data instead of one flat threshold applied to every server type.

- To train, compare and honestly report the accuracy of machine learning models that estimate CPU utilization and power draw from environmental and workload features, used as an explainable validation layer.
- To forecast near-term server workload at multiple horizons, using whichever method fits each horizon rather than forcing a single model onto every time range.
- To simulate, before any recommendation is shown, whether consolidating an idle server onto a candidate target would be safe both right now and in the near future, across CPU, memory, network and thermal dimensions.
- To rank pending recommendations using a configurable, weighted combination of estimated energy, cost, carbon and water impact against operational risk.
- To keep the human operator as the sole decision-maker: every decision and its supporting estimate is permanently logged for audit and ESG reporting and role-based access control separates who may decide from who may attest execution.

1.4 SCOPE OF THE PROJECT

The current implementation covers the complete pipeline from telemetry ingestion through to ranked, human approved recommendations and an ESG decision log. The system runs against a simulated fleet of thirteen servers spanning four server types (Compute, GPU, Storage and Edge), three cooling types (Air, Hybrid and Liquid) and three regions (APAC, EU and Middle East), with several servers in each type deliberately biased toward lower utilization so the recommendation engine always has more than one genuine candidate to work with.

Telemetry for all four domains (compute, power, cooling and storage) is produced by purpose built simulators rather than read from physical hardware, since production data center access was not available for this project. These simulators were built to mirror real monitoring behavior closely, including realistic posting intervals, range validated readings and duplicate delivery handling, so that the ingestion and analysis layers behave exactly as they would against a live facility. The two estimation models (CPU and power) are trained offline on a real, static 10,000 row “Green AI Data Center Telemetry” dataset. The two forecasting models (CPU and memory) are trained on telemetry accumulated by the live simulator itself.

The project covers rule based waste detection across all four categories (idle servers, stale data, duplicated data and over provisioned storage), with per type thresholds derived statistically. It covers a four dimension consolidation safety check spanning CPU, memory, network and thermal headroom, evaluated both against a server's current state and its near term forecast. It covers a priority weighted recommendation engine whose scoring weights an operator can adjust live and a role based access system that separates who may approve a recommendation from who may later confirm it was carried out. It also covers a dedicated ESG reporting layer that aggregates approved decisions into a facility level sustainability record, downloadable as CSV or PDF.

Cost and carbon figures throughout the system are computed using placeholder tariff and grid intensity constants, whether it is a predicted power draw, an expected saving or a water usage figure, is clearly distinguished in the interface from a value that was actually measured. That distinction was treated as a core design requirement throughout the project, since a system meant to eventually support real sustainability reporting has to be honest about which of its numbers can be trusted at face value and which are a considered estimate.

CHAPTER 2

LITERATURE SURVEY

Research on data center efficiency has developed along a few separate paths that rarely meet. One path optimizes where workloads run. Another applies machine learning to predict resource use before or after the fact. A third looks at carbon-aware scheduling and cooling. A fourth is less concerned with optimization and asks instead whether sustainability is even being measured and reported at all. The study of these existing approaches provides valuable insights into the development of the proposed GreenOps system.

2.1 LITERATURE REVIEW

Ahmed, Brown and Gegov investigated dynamic resource allocation through workload prediction in “Dynamic Resource Allocation through Workload Prediction for Energy-Efficient Computing” [1]. Their work examined how workload prediction can support dynamic resource allocation and improve energy efficiency while maintaining service requirements. This approach demonstrates the value of predictive resource management. However, it does not combine workload prediction with a unified set of sustainability metrics, recommendation generation, what-if simulation and human approval.

Alex, Ojo and Awuor proposed a carbon-aware and energy-efficient virtual machine placement approach using deep Q-networks and agglomerative clustering [2]. Their work considers carbon emissions, energy efficiency and service-level agreement requirements when determining virtual machine placement. Although this approach supports sustainability-oriented resource allocation, its primary focus is placement optimization rather than an integrated workflow for telemetry monitoring, waste detection, forecasting, recommendations and operator decision-making.

Bawa, Kaur Chahal and Kaur proposed an ensemble learning approach for cloud resource management and workload prediction using XGBoost and LightGBM [3]. Their study investigated workload activity classification and the prediction of CPU and memory utilization. The findings highlight the usefulness of ensemble learning for anticipating resource requirements. However, prediction remains the primary outcome and is not directly connected to a sustainability-aware recommendation mechanism for selecting corrective actions.

Beloglazov, Abawajy and Buyya, in “Energy-Aware Resource Allocation Heuristics for Efficient Management of Data Centers for Cloud Computing,” proposed heuristics for virtual machine placement and migration to improve data center energy efficiency [4]. Their work demonstrated the value of dynamic consolidation for reducing power consumption compared with static resource allocation. However, the approach focuses on automated resource management and does not provide the same operator-reviewed recommendation workflow proposed in GreenOps.

Deva and Jain investigated cloud resource utilization prediction using Azure Automated Machine Learning and XGBoost in enterprise environments [5]. Their work explores automated model selection and machine-learning-based prediction of cloud resource utilization. This demonstrates how machine learning can support resource planning and utilization analysis. However, resource prediction alone does not provide a complete workflow for assessing sustainability impacts, recommending corrective actions and obtaining operator approval.

The European Commission has established corporate sustainability reporting requirements that require applicable organizations to disclose environmental and sustainability-related information [6]. These requirements emphasize the

importance of systematic sustainability measurement and transparent reporting. However, reporting frameworks primarily address disclosure obligations rather than providing continuous infrastructure monitoring, machine-learning-based prediction or operational recommendations.

Gorey discussed the land and water implications associated with the rapid expansion of artificial intelligence infrastructure [7]. The discussion highlights how the growth of data centers can place additional pressure on natural resources and the regions hosting these facilities. This broader environmental perspective shows why data center sustainability should not be assessed through electricity consumption alone. However, the work does not propose an operational mechanism for continuous resource monitoring, prediction or optimization.

Grochulska-Salak et al. examined sustainable data centers through the use of renewable energy and green-blue infrastructure [9]. Their work considers several sustainability dimensions, including energy efficiency, water usage, waste heat recovery, renewable energy integration and land use. This demonstrates the importance of assessing data center sustainability through multiple indicators. However, a broad sustainability framework does not itself implement a continuous machine-learning-based system for detecting inefficiencies and recommending operational changes.

Khan, Tian, Ilager and Buyya investigated workload forecasting and energy-state estimation in cloud data centers using multiple resource characteristics, including provisioned resources, disk throughput and network activity [10]. Their work highlights the importance of considering different resource parameters when estimating workload behaviour and energy-related characteristics. However, forecasting and estimation do not by themselves

provide an integrated workflow for waste detection, sustainability assessment, recommendation generation and operator approval.

Lovász, Niedermeier and de Meer studied the performance implications of energy-aware virtual machine consolidation in “Performance Tradeoffs of Energy-Aware Virtual Machine Consolidation” [11]. Their work considered the trade-off between energy savings and system performance, showing that consolidation should not be evaluated only in terms of energy reduction. This is important because aggressive consolidation can affect performance. However, the study primarily focuses on consolidation trade-offs rather than a continuous system that combines forecasting, sustainability assessment and operator-reviewed recommendations.

Radovanović et al. presented “Carbon-Aware Computing for Datacenters,” which describes an approach for shifting flexible workloads across time and geographical locations according to forecasted grid carbon intensity [13]. This work demonstrates how carbon emissions can be considered alongside computing requirements when scheduling workloads. However, this approach depends on workload flexibility and access to infrastructure across different locations, which may not be available in smaller or single-site data center environments.

Subramanian, Vasan and Sivasubramaniam investigated server consolidation in “Reducing Data Center Power with Server Consolidation: Approximation and Evaluation” [17]. They treated server consolidation as an optimization problem similar to bin packing, aiming to reduce power consumption while satisfying resource requirements. However, the approach does not integrate machine-learning-based workload forecasting with broader sustainability metrics and operator-oriented decision support.

Tao and Gao examined the environmental and public-health implications associated with global data center expansion and highlighted the need for further empirical investigation into these impacts [18]. Their work broadens the discussion of data center sustainability beyond operational energy consumption to include potential environmental and societal consequences. However, the study focuses on broader environmental impacts rather than providing a practical system for operational resource optimization and machine-learning-based decision support.

Tripathi et al., in research conducted by the Council on Energy, Environment and Water and SYSTEMIQ, examined the development of India's data center ecosystem and its associated infrastructure, energy, resource and resilience requirements [19]. Their work highlights the importance of planning for the growing demands of data center infrastructure. However, ecosystem-level analysis does not directly provide an operational system for detecting resource inefficiencies and recommending corrective actions within individual data centers.

Uddin et al. investigated server consolidation as a technique for improving cloud data center power efficiency and reducing overall cost of ownership [20]. Their work examines the relationship between consolidation, power consumption and operational costs. However, the primary focus is on evaluating consolidation strategies rather than continuously detecting inefficiencies, forecasting future demand and guiding operators through a structured decision-making process.

Vaccaro, Tagliabue and Aldinucci investigated sustainable data centers with an emphasis on energy efficiency and resource optimization [21]. Their work contributes to the broader study of improving resource utilization and sustainability in data center environments. However, the study's broad

sustainability and optimization perspective does not itself provide the complete integrated monitoring-to-decision workflow proposed in GreenOps.

Zhan et al. investigated data center cooling system optimization using offline reinforcement learning [23]. Their approach uses historical operational data to learn cooling-control policies without relying on direct trial-and-error experimentation on live infrastructure during training. The study demonstrates the potential of reinforcement learning for cooling optimization, including its application in a production data center. However, its primary focus is cooling control rather than combining workload forecasting, resource waste detection, multiple sustainability metrics and human-reviewed recommendations in one system.

From the reviewed literature, several research directions can be identified. Existing studies have addressed workload prediction and intelligent resource allocation [1]–[5], sustainability measurement and environmental impact [6], [7], [9], [18], [19], workload forecasting and energy-state estimation [10], energy-aware consolidation [11], [17], [20], carbon-aware scheduling [13], and sustainable resource optimization [21], [23]. However, these capabilities are generally addressed separately.

The major research gap identified from the literature is the limited integration of continuous telemetry monitoring, rule-based waste detection, machine-learning-based prediction, sustainability assessment, recommendation generation, what-if simulation and human-reviewed decision-making within a unified operational workflow. Existing approaches often focus on a particular optimization task, prediction model or sustainability assessment. GreenOps addresses this integration gap by connecting these capabilities while retaining the final operational decision with the human operator.

Thus, the reviewed literature establishes the importance of intelligent resource management, predictive modelling and comprehensive sustainability measurement in data centers. GreenOps builds on these research directions by combining waste detection, prediction, sustainability assessment, recommendation generation and what-if simulation into a unified decision-support platform.

2.2 SUMMARY OF LITERATURE SURVEY

Table 2.1 Comparative Analysis of Existing Literature

Ref.	Study	ML / Optimization Technique Used	How the Model / Technique Was Used	Main Focus	Limitation / Gap Relative to GreenOps
[1]	Ahmed et al.	Neural-network-based workload prediction	Predicts workload behaviour to support dynamic resource allocation and improve energy efficiency.	Energy-efficient dynamic resource allocation	Does not combine prediction with unified sustainability metrics, recommendations, what-if analysis and human approval.
[2]	Alex et al.	Deep Q-Network and Agglomerative Clustering	Uses clustering and deep reinforcement learning to guide carbon-aware and energy-efficient VM placement.	Carbon-aware VM placement	Focuses on VM-placement optimization rather than an integrated monitoring, forecasting, recommendation and human-decision workflow.
[3]	Bawa et al.	XGBoost and LightGBM ensemble learning	Uses ensemble models to identify workload activity and predict CPU and memory utilization.	Workload prediction and cloud resource management	Prediction is the main outcome. It does not connect predictions to sustainability-aware recommendations or what-if analysis.

[4]	Beloglazov et al.	Adaptive VM-consolidation heuristics	Detects overloaded and underloaded hosts, migrates VMs, consolidates workloads and powers down idle servers.	Energy-aware VM allocation and server consolidation	Does not provide the integrated forecasting, sustainability assessment, what-if analysis and human-reviewed recommendation workflow proposed in GreenOps.
[5]	Deva and Jain	Azure AutoML and XGBoost	Uses Azure AutoML for model selection and XGBoost for predicting cloud resource utilization.	Cloud resource-utilization prediction	Focuses on prediction and does not extend the results into integrated sustainability assessment, recommendations and human-reviewed decisions.
[6]	European Commission	Sustainability reporting framework	Establishes requirements for reporting environmental and sustainability-related information.	Corporate sustainability reporting	Focuses on disclosure and reporting rather than real-time monitoring, prediction or operational recommendations.
[7]	Gorey	Environmental and resource-impact analysis	Examines land and water implications associated with the expansion of AI infrastructure.	Land and water impact of AI infrastructure	Highlights resource impacts but does not provide an operational mechanism for monitoring, prediction or optimization.
[9]	Grochulska-Salak et al.	Sustainability indicators and framework	Considers energy efficiency, renewable energy, water use, waste heat, carbon emissions and land use.	Holistic data center sustainability	Provides a broad sustainability perspective but does not implement a continuous ML-based operational optimization system.

[10]	Khan et al.	Linear Regression, Ridge, ARD, ElasticNet, GRU and clustering	Uses forecasting models for workload prediction and clustering for VM energy-state estimation.	Workload forecasting and energy-state estimation	Does not integrate forecasting with waste detection, unified sustainability metrics, recommendations, what-if simulation and operator approval.
[11]	Lovász et al.	Energy- and performance-aware consolidation algorithms	Evaluates VM consolidation while considering both energy savings and possible performance degradation.	Energy versus performance trade-off	Focuses on consolidation trade-offs rather than continuous prediction, sustainability assessment or recommendation generation.
[13]	Radovano vić et al.	Carbon-intensity forecasting, demand prediction and risk-aware optimization	Uses forecasts and optimization to shift flexible workloads toward lower-carbon periods and locations.	Carbon-aware workload scheduling	Depends on workload flexibility and distributed infrastructure. Does not focus on local resource waste detection or human-reviewed recommendations.
[17]	Subramanian et al.	Approximation algorithm	Models server consolidation as a variable-sized bin-packing problem to reduce the number of active servers and power consumption.	Power reduction through server consolidation	Does not use ML forecasting or integrate broader sustainability metrics with operator decision support.
[18]	Tao and Gao	Environmental impact analysis	Examines environmental and public-health implications associated with data center expansion.	Environmental and societal impact of data centers	Focuses on broader environmental impacts rather than operational resource optimization or ML-based decision support.

[19]	Tripathi et al.	Industry and sustainability analysis	Examines infrastructure, energy, resource and resilience requirements associated with India's growing data center ecosystem.	Data center infrastructure , energy and resilience	Provides ecosystem-level analysis rather than an operational system for detecting and resolving resource inefficiencies.
[20]	Uddin et al.	Server-consolidation techniques	Evaluates consolidation strategies and their effects on power efficiency and total cost of ownership.	Energy consumption, power efficiency and operational cost	Primarily evaluates consolidation outcomes rather than providing a continuous detection, prediction, recommendation and decision workflow.
[21]	Vaccaro et al.	Sustainability and resource-optimization approaches	Examines approaches for improving energy efficiency and resource utilization in sustainable data centers.	Sustainable data center operation	Does not implement the complete ML-based monitoring-to-decision workflow proposed in GreenOps.
[23]	Zhan et al.	Physics-informed offline reinforcement learning	Learns cooling-control policies from historical operational data and applies the learned policy to optimize cooling operations.	Data center cooling optimization	Focuses specifically on cooling control rather than integrating workload forecasting, resource waste detection, broader sustainability metrics and human-reviewed recommendations.

From the comparative analysis, existing studies primarily focus on individual aspects of sustainable data-center management such as resource allocation, server consolidation, workload prediction, carbon-aware scheduling, cooling optimization or sustainability measurement. Although several studies employ

machine learning or intelligent optimization techniques, these capabilities are generally addressed independently. Limited attention is given to integrating continuous telemetry monitoring, waste detection, workload forecasting, sustainability assessment, recommendation generation and what-if simulation into a single operator-oriented workflow. GreenOps addresses this gap by connecting these capabilities while retaining human oversight over the final operational decision.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 OVERVIEW OF THE PROPOSED SYSTEM

GreenOps watches four streams of data center telemetry, uses simple explainable rules to spot waste, uses two families of machine learning models to forecast and validate server behaviour, simulates the consequence of consolidating workloads before anything happens, and hands a human operator a ranked list of recommendations with honest, itemized impact estimates. The system never executes anything itself.

3.2 SYSTEM ARCHITECTURE

The architecture is organized into four layers: a data layer of independent telemetry simulators, a FastAPI backend that performs ingestion, storage, metrics calculation, rule-based waste detection, ML validation, forecasting, what-if simulation, and recommendation ranking; a PostgreSQL persistence layer; and a React frontend that exposes the results through role-appropriate dashboards. Figure 3.1 shows the complete architecture, from the four simulated monitors on the left, through the backend's seven internal stages, to the frontend pages on the right and the database tables at the bottom.

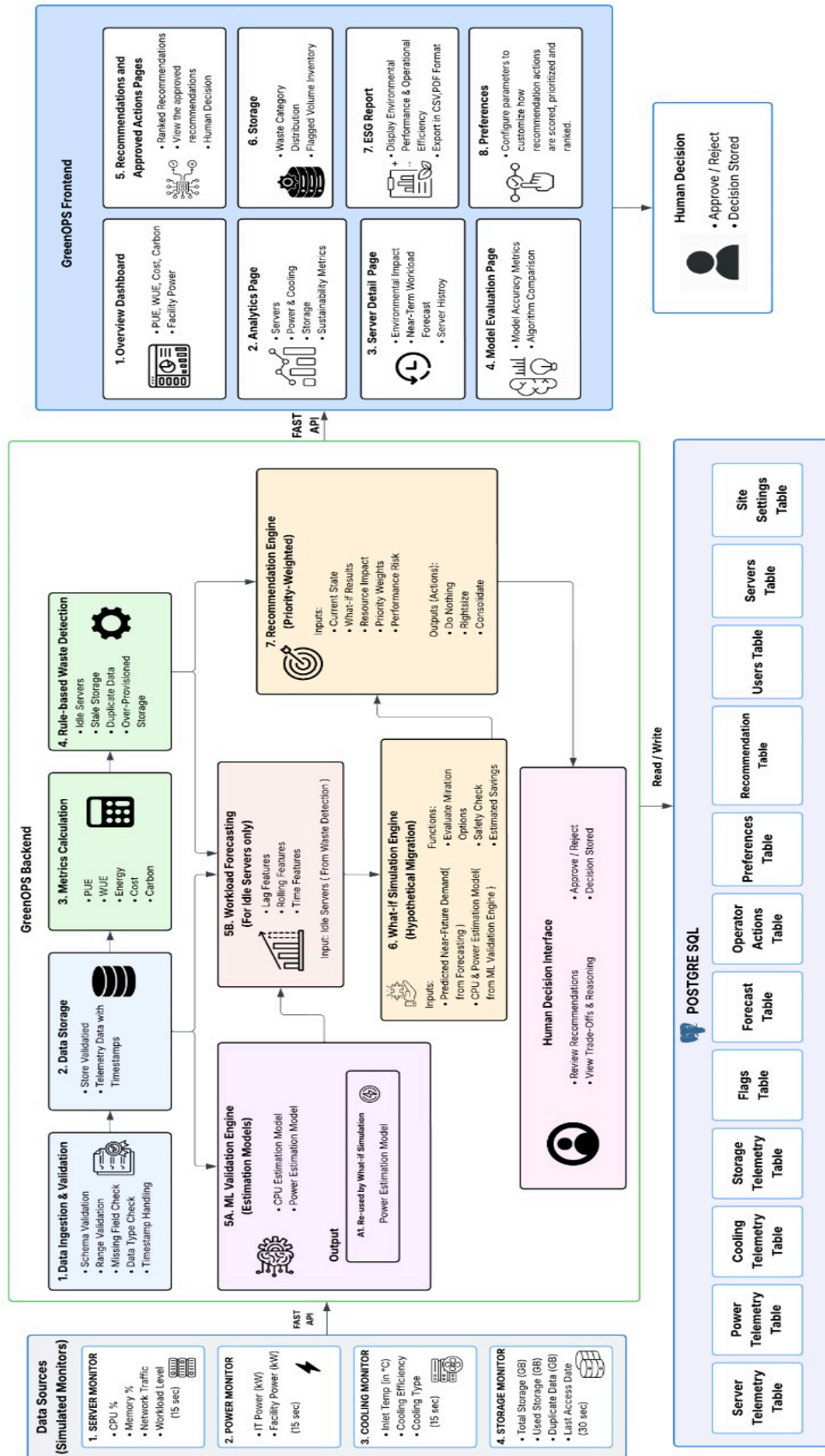


Fig 3.1 GreenOps System Architecture

At the center of the backend, the Rule-based Waste Detection module and the Metrics Calculation module both read from validated, stored telemetry and feed two parallel downstream paths: an ML Validation Engine (offline-trained CPU and power estimation models) and a Workload Forecasting module restricted to servers already flagged idle. Both paths feed a What-if Simulation Engine, which in turn feeds the priority-weighted Recommendation Engine. Every recommendation the engine produces is surfaced to a Human Decision Interface, and the operator's approve/reject/snooze decision is written back to the database, closing the loop.

3.3 SYSTEM DESIGN

3.3.1 ACTIVITY DIAGRAM

The activity diagram shown in Figure 3.2 illustrates the continuous monitoring and analysis workflow executed by the GreenOps telemetry pipeline. The process initiates with the telemetry cycle, where independent simulator scripts post structured readings every 15 to 30 seconds. Upon ingestion, the FastAPI backend performs schema validation and range checks. To ensure data integrity, any duplicate payload received within a 2-second sliding window is automatically dropped and marked as ignored.

Validated telemetry flows into the rule-based waste detection engine, which evaluates the latest observations against domain-specific thresholds for idle servers, stale data, duplicate data, and over-provisioned storage. If no waste condition is met, any previously active flag for that condition is resolved. When a condition is triggered, a new flag is raised or updated. For non-compute waste flags (archive, deduplicate, rightsize), recommendations are directly marked as pending. For idle server flags, the pipeline initiates a 15-minute workload forecast for the source server. If a safe candidate target exists and the source server is projected to remain idle, a consolidation recommendation is queued for

operator review; otherwise, the recommendation is temporarily blocked and re-evaluated during the next cycle.

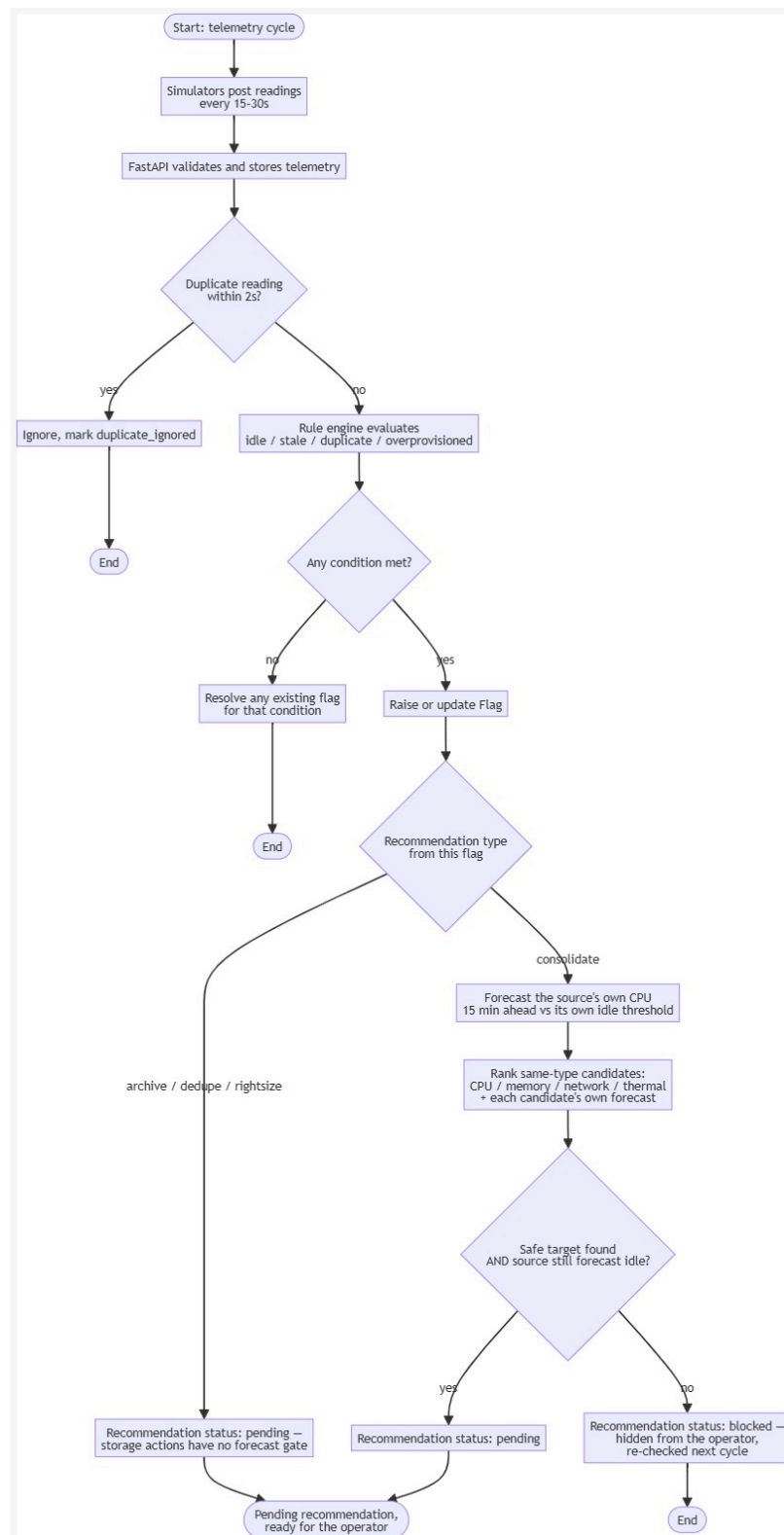


Fig 3.2 Activity Diagram for Monitoring and Analysis

Figure 3.3 details the decision-making and execution workflow for pending recommendations. Pending actions - already validated for safety in the case of server consolidations - are first prioritized and ranked by the recommendation engine using the system's weighted scoring matrix across energy, carbon, cost, water, and operational risk metrics. The ranked list is then presented to the operator through the Recommendations dashboard interface.

The human operator evaluates the presented options and selects one of three primary decision paths: Accept, Snooze, or Dismiss. Accepting a recommendation records an OperatorAction audit log entry, freezing a snapshot of the estimated savings at decision time. The accepted action can later be manually attested as executed on the Approved Actions page, allowing the ESG and compliance modules to aggregate the verified decision history. Selecting Snooze assigns a deferred timestamp, after which the recommendation automatically reverts to pending status. Dismissing a recommendation records a do-nothing action, concluding the lifecycle for that specific evaluation cycle.

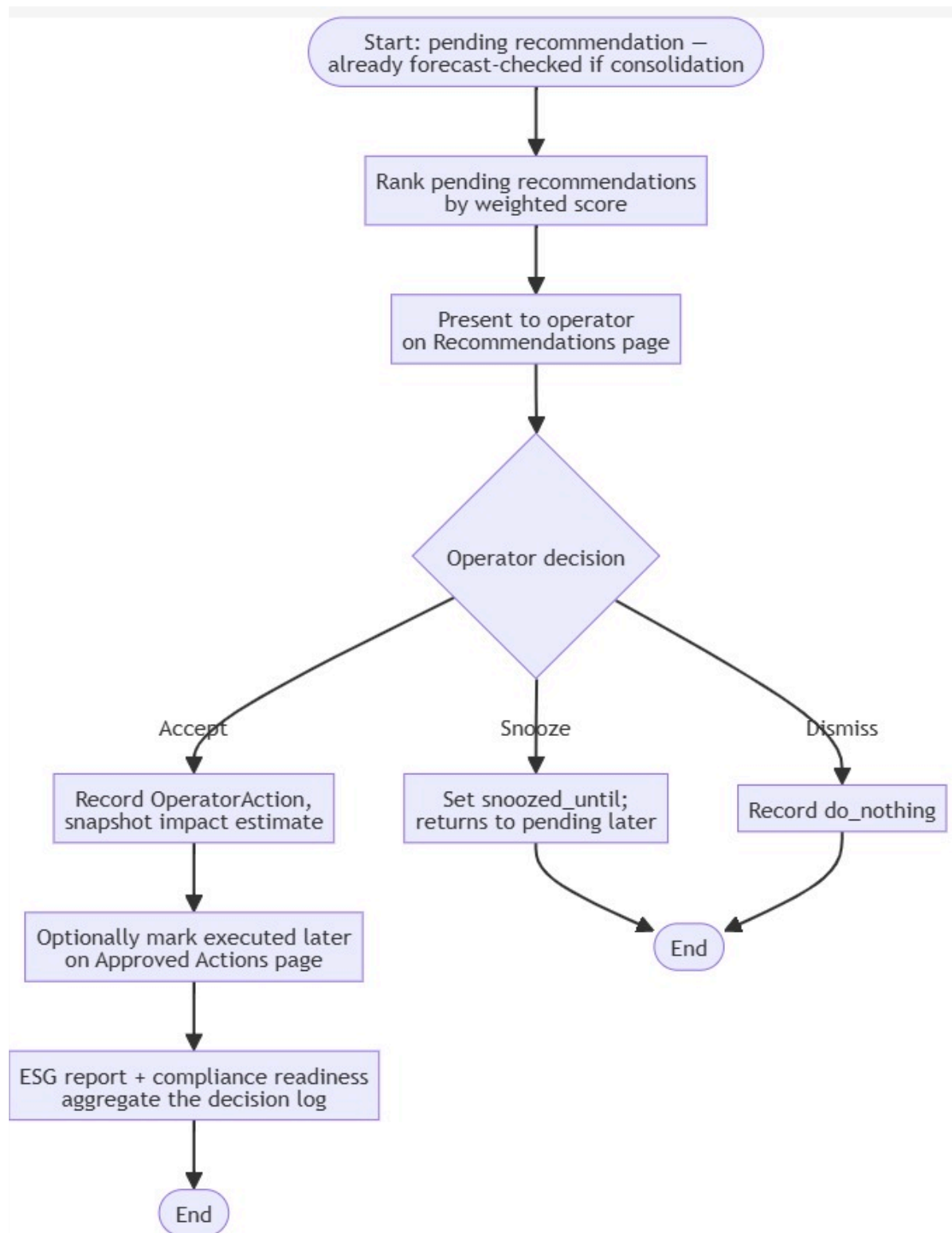


Fig 3.3 Activity Diagram for Recommendation and Action

3.3.2 ER DIAGRAM

Figure 3.4 shows the Entity-Relationship (ER) diagram for the GreenOps platform, which maps out the core database tables and how they connect. The **SERVER** entity sits at the center, holding each server's identifier, type, cooling type and datacenter region, and every other operational table points back to it through `server_id`. Each server posts readings to four telemetry tables: **SERVER_TELEMETRY** for CPU, memory and network metrics, **POWER_TELEMETRY** for IT and facility power, **COOLING_TELEMETRY** for inlet temperature and cooling efficiency, and **STORAGE_TELEMETRY** for total, used and duplicate data volumes. These links are one-to-many and optional, so a server can be registered before any data arrives and then build up a time-stamped history. The **FORECAST** table also ties to **SERVER**, storing predicted CPU values for a chosen horizon next to the measured ones.

The rule engine reads this telemetry and raises **FLAG** entries against the affected server, each with a type, severity, metric value and resolution state. Every flag produces a **RECOMMENDATION** with a priority, a workflow status and an optional snooze date, ranked using the energy, cost, carbon, water and risk weights in **PREFERENCES**. **SITE_SETTINGS** supplies the wider context, such as the grid emission factor, installed IT capacity and renewable energy factor, which feed the estimates attached to each decision. Operators, stored in the **USER** entity, review each recommendation and choose what to do. Each decision is saved in **OPERATOR_ACTION** with the action taken, the servers involved, estimated savings in energy, carbon and cost, and the deciding user, giving a full audit trail from raw data to the final human decision.

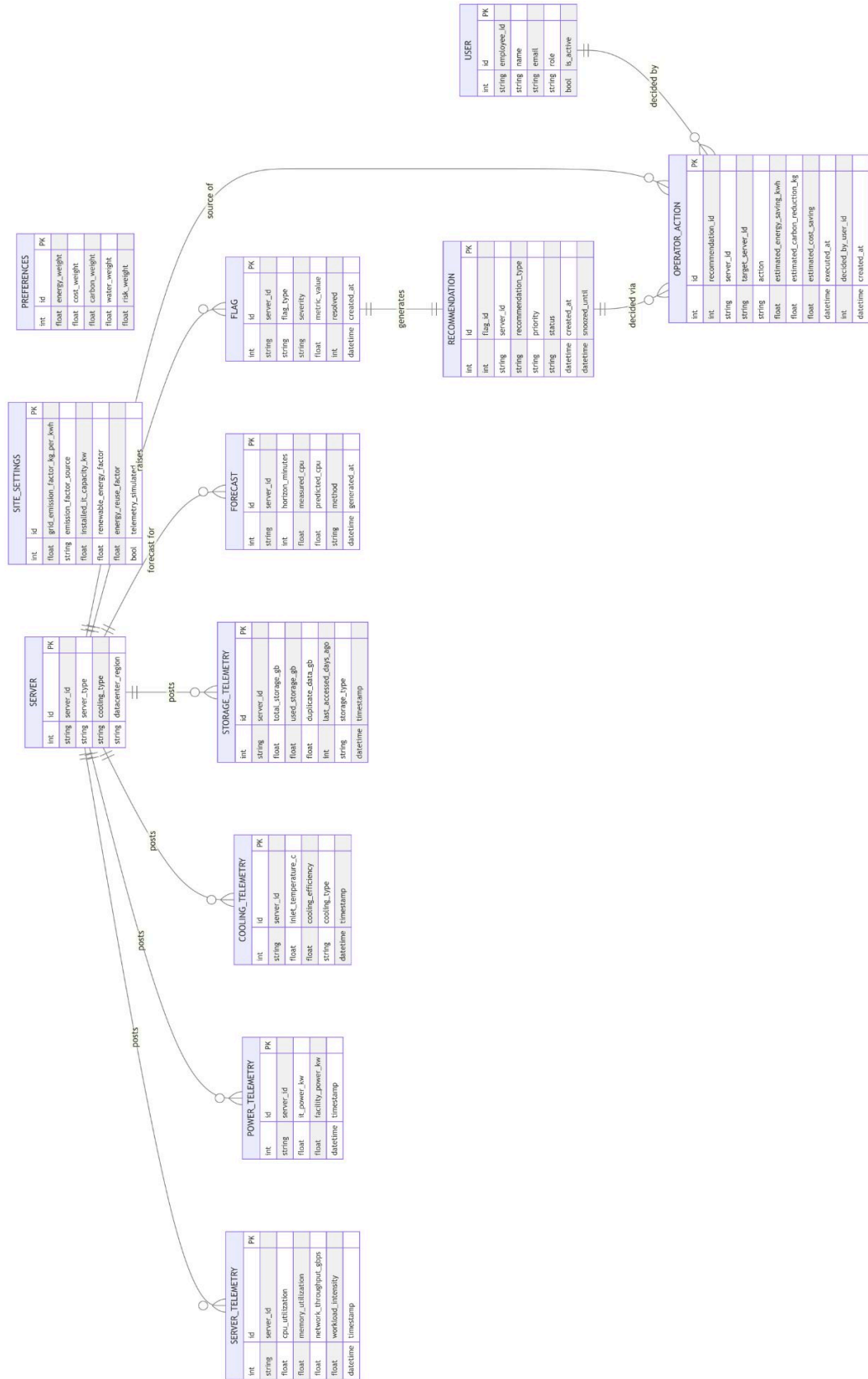


Fig 3.4 ER Diagram

3.4 DATA LAYER - TELEMETRY SIMULATION

Nothing in the running system reads real hardware. Four independent Python scripts each generate one telemetry domain and POST it to the FastAPI backend every fifteen seconds, per server, continuously, standing in for real monitoring integrations that were not available for this project.

Table 3.1 Telemetry Simulators and Their Output Fields

Simulator	Posts to	Fields
server_monitor.py	/telemetry/server	cpu_utilization, memory_utilization, network_throughput_gbps, workload_intensity
power_monitor.py	/telemetry/power	it_power_kw, facility_power_kw
cooling_monitor.py	/telemetry/cooling	inlet_temperature_c, cooling_efficiency, cooling_type
storage_monitor.py	/telemetry/storage	total_storage_gb, used_storage_gb, duplicate_data_gb, last_accessed_days_ago

Thirteen simulated servers (S1–S13) exist, each with a fixed identity assigned at registration and never changed at runtime, spanning four server types, three cooling types, and three regions, with a deliberately varied workload bias so that some servers are realistically idle and others realistically busy.

Table 3.2 Simulated Server Identity and Workload Bias

Server	Type	Cooling	Region	Workload bias
S1	Compute	Air	APAC	0.65
S2	Compute	Air	APAC	0.60
S3	GPU	Hybrid	EU	0.30
S4	Storage	Liquid	ME	0.15 (idle)
S5	Edge	Air	APAC	0.12 (idle)
S6	Compute	Air	APAC	0.05 (very idle)
S7	GPU	Hybrid	EU	0.55
S8	GPU	Hybrid	ME	0.10 (idle)

Server	Type	Cooling	Region	Workload bias
S9	Storage	Liquid	ME	0.45
S10	Storage	Liquid	APAC	0.14 (idle)
S11	Edge	Air	APAC	0.50
S12	Edge	Air	EU	0.16 (idle)
S13	Compute	Air	ME	0.70

Each simulator generates its readings from a small, explicit formula rather than pure random noise, so that later modules can be verified by hand. `server_monitor.py` draws a workload value around each server's fixed bias, and derives CPU, memory, and network from it, so memory and network carry no information CPU does not already contain. `power_monitor.py` reads each server's live CPU and memory over the API before generating its own reading, so that IT power is a real function of load, and the ratio of facility power to IT power (PUE) is a real function of the server's live cooling efficiency-both of these were previously pure random draws and were fixed during this project. `cooling_monitor.py` similarly reads each server's live CPU and applies a per-cooling-type temperature-rise and efficiency-drop curve, so that air-cooled servers heat up fastest and lose efficiency fastest under load, hybrid sits between, and liquid barely moves; this causal direction runs only from load to environment, never the reverse, which is a known and explicitly documented limitation. `storage_monitor.py` generates capacity, staleness, and duplication metadata.

FastAPI validates every incoming reading against a Pydantic schema, writes it to the database with a timestamp, and silently drops an exact-duplicate delivery from the same server arriving within two seconds, so that a network retry never double-counts a reading.

3.5 RULE-BASED WASTE DETECTION ENGINE

Four checks, each simple enough to verify by hand, run continuously against the latest stored telemetry.

Table 3.3 Rule Engine Flags

Flag	Rule
idle_server	6-hour average CPU below that server type's threshold
stale_data	last_accessed_days_ago > 90
duplicate_data	duplicate_data_gb ÷ total_storage_gb > 20%
overprovisioned	used_storage_gb ÷ total_storage_gb < 30%

The idle threshold is not a single flat number applied to every server. Per-type thresholds are derived once from the real “**Green AI Data Center Telemetry**” dataset as

$$\text{threshold} = \text{mean_cpu}(\text{type}) - 1.5 * \text{std_cpu}(\text{type})$$

clamped to the range [5%, 35%], with a severity weight scaled to that type's average power draw relative to the lowest-power type. This means an idle GPU server, which draws roughly 66% more power on average than an idle Edge server at the same CPU percentage in the real dataset, is flagged as materially higher-priority waste rather than identically to any other idle server.

Table 3.4 Per-Type Idle Thresholds and Severity Weights

Server type	Idle threshold	Severity weight
GPU	16.9%	1.66×
Compute	17.3%	1.22×
Storage	16.8%	1.06×
Edge	18.9%	1.0× (baseline)

A flag that stops being true is resolved on the same cadence it was raised, so nothing accumulates indefinitely; the synchronization routine also withdraws any pending recommendation whose underlying flag is no longer active, so a server flapping between idle and healthy does not pile up stale duplicate recommendations.

3.6 SUSTAINABILITY METRICS

Table 3.5 lists every sustainability metric the system computes, together with an honest statement of how trustworthy each one is—a distinction the system is deliberately explicit about throughout, rather than presenting every number with equal apparent confidence.

Table 3.5 Sustainability Metrics and Trust Level

Metric	Formula	Trust level
PUE	$\text{facility_power_kw} \div \text{it_power_kw}$	Genuine ratio of two directly simulated values
Energy (kWh)	$\Sigma (\text{Power} * \text{Elapsed Time})$	Corrected; previously over-counted $\approx 4\times$ by assuming one reading per minute
Cost	$\text{energy_kwh} \times \text{Tariff}$	Estimate
Carbon	$\text{energy_kwh} \times \text{Carbon Factor}$	Estimate
Total water (L)	Cumulative $\text{energy_kwh} \times \text{WUE factor}$ by cooling type, since telemetry began	Estimate, running total
WUE (L/kWh)	$\text{Water Consumption} \div \text{Energy}$	Estimate, genuine ratio in shape

Water and carbon figures use published industry-average placeholders (WUE factors of 0.3, 1.8, and 0.9 L/kWh for Air, Hybrid, and Liquid cooling respectively) keyed only by the categorical cooling type, never by the live numeric cooling-efficiency reading. In a production deployment, cost, carbon, and WUE would be replaced by a real electricity tariff, a live grid carbon-intensity feed, and a real facility water meter.

3.7 MACHINE LEARNING VALIDATION MODELS

Two regression models are trained once, offline, on a real, static, 10,000-row Kaggle dataset (“Green AI Data Center Telemetry”)-never on live simulator output, so the two pipelines share only a schema, never an origin. Shared preprocessing scales numeric features and one-hot-encodes categorical features with the first category dropped, which is necessary because, without dropping it, one-hot-encoding every category alongside an intercept term creates a perfectly collinear feature matrix, a bug that was found and fixed during development.

3.7.1 CPU ESTIMATION MODEL

Target: `cpu_utilization`.

Inputs: `workload_intensity`, `memory_utilization`, `network_throughput_gbps`, `inlet_temperature_c`, `cooling_efficiency`, `pue`, plus the categorical features `server_type`, `cooling_type`, `datacenter_region`, and `time_of_day`.

The model deliberately excludes its own target from its inputs to avoid leakage.

3.7.2 POWER ESTIMATION MODEL

Target: power, rescaled from the dataset's 28–233 kW range down to the live simulated 1.2–4.5 kW range-a pure unit change, confirmed by identical R^2 before and after rescaling.

Inputs are the same six numeric and four categorical features as the CPU model, plus `cpu_utilization`, which is legitimate here since CPU is not the target for this model.

3.7.3 MODEL SELECTION RULE

Three candidate algorithms-Linear Regression, Random Forest (200 estimators), and XGBoost (200 estimators, depth 5) are trained for each target on an 80/20 split. The lowest mean absolute error wins by default, but a more complex

model (Random Forest or XGBoost) must beat Linear Regression by at least 5% relative MAE improvement to be selected over it, a promotion bar that was fixed before results were seen. On this dataset, Linear Regression won both targets detailed results are reported in Chapter 5.

3.8 WORKLOAD FORECASTING

The forecasting module answers one question per tier: given a server's history, what will its CPU likely be? Two structurally different forecasting jobs exist and are not interchangeable.

3.8.1 *OPERATOR-FACING FORECAST*

Gated to servers currently flagged idle, this forecast covers three horizons, each handled by the method appropriate to its range rather than one method forced onto all three. The one-hour horizon uses its own trained lagged-feature model (five-, ten-, fifteen-, and twenty-minute CPU lags, the rolling one-hour average, memory, workload intensity, network throughput, and time-of-day encoded as sine/cosine), falling back to short-horizon trend extrapolation until at least thirty real lagged samples exist. The six-hour and twenty-four-hour horizons deliberately use no trained model at all: there is not yet enough real history at that range, and because each server's workload is generated from a fixed bias that never drifts or cycles, the honest estimate that far out is reversion to the server's own recent average rather than a straight-line extrapolation, which would badly overshoot. The long-horizon forecast explicitly refuses to answer - returning **“not enough history”** rather than a guess-unless real telemetry coverage reaches at least half the requested window with at least ten samples.

3.8.2 *CONSOLIDATION SAFETY FORECAST*

A completely separate, fifteen-minute-horizon model, used internally only by the what-if simulation described in Section 3.8, and always run on any

registered server with no idle gate-since a consolidation target is, by definition, not idle, and could never pass an idle-only gate. This forecast reuses the same CPU-lag architecture as the operator-facing model, plus a second, independently trained memory forecast model, since the safety check needs both dimensions.

Live prediction errors for any of these models will typically run higher than the reported offline metrics, because the models were trained on real historical relationships while live data comes from simulators using simpler generating formulas-a genuine, expected train/inference gap rather than a defect.

3.9 RECOMMENDATION ENGINE AND WHAT-IF SIMULATION

Each active flag is mapped to a recommended action type.

Table 3.6 Flag-to-Recommendation Mapping

Flag	Recommended action
idle_server	consolidate
stale_data	archive
duplicate_data	deduplicate
overprovisioned	rightsize

For idle_server flags specifically, an actual what-if simulation is run rather than a source-only guess. Every same-type server that is not itself idle-flagged is considered a candidate. For each candidate, post-move CPU, memory, and network are computed by adding the source server's own load, discounted by a 0.9 overhead factor, onto the candidate's current load. Each candidate is then checked across four safety dimensions, all of which must pass for the candidate to be considered safe right now.

Table 3.7 Consolidation Safety Check Dimensions

Dimension	Limit	Basis
CPU	< 75%	Safety limit constant
Memory	< 75%	Safety limit constant
Network	< 18.75 Gbps	75% of an assumed 25 Gbps NIC
Thermal	Predicted post-move inlet temp < 30°C	Reuses the cooling simulator's own load→temperature formula

Independently of the current-state check, the candidate's own near-term CPU and memory forecast (15 minutes ahead, Section 3.7.2) is checked against the same limits. A candidate only counts as fully safe when both the current-state check and the forecast check pass - this catches a target that looks fine right now but is about to get busy on its own. Network and thermal have no forecast counterpart, since no trained model exists for either, so they remain current-snapshot checks only. The top-ranked safe candidate (safe first, then most remaining headroom) becomes the recommended target; if no candidate qualifies, the recommendation is explicitly marked unsafe with a stated reason, never silently approved, and all considered candidates (up to five) are exposed in the response so an operator can see what else was evaluated. The impact estimate for a recommended consolidation prefers the trained power model's prediction for the target's full post-move profile, scaled to facility power by the target's current PUE, falling back to a simpler CPU-ratio formula only if the model or the target's cooling telemetry is unavailable; every use of the power model in an estimate is stated explicitly, including the caveat that it is a legitimate estimate rather than a measurement.

Pending recommendations across all flag types are ranked by:

$$\text{score} = \sum(\text{weight}_i \times \text{normalized_impact}_i) - \text{risk_weight} * \text{risk}$$

normalized against the maximum of each dimension within the current pending batch so that kWh, currency, kilograms, and litres do not dominate each other purely by raw magnitude. The weights are stored in a single-row, organization-configurable Preferences table editable from the frontend; changing them re-ranks recommendations instantly without retraining anything. Risk for a consolidation recommendation is

$$\text{max(post_move_cpu, post_move_memory)} \div 75$$

for storage-only actions, which have no target server and no power effect, risk is a fixed 0.1.

3.10 OPERATOR DECISIONS AND AUDIT TRAIL

An operator with the `infrastructure_manager` role can consolidate (or the generic `rightsize` verb, for archive/deduplicate actions), snooze, or dismiss any pending recommendation. Every decision snapshots the what-if impact exactly as it was shown at the moment of the decision into an `OperatorAction` record-the ESG report reads this frozen snapshot, not a live recalculation, so the log always reflects exactly what the operator actually saw when deciding. Snoozing sets a `snoozed_until` timestamp; once that passes, the recommendation quietly returns to pending rather than being lost.

The `Approved Actions` page is the one narrow exception to the rule that GreenOps has no way to verify anything actually happened: an operator can manually attest, after the fact, that a consolidation was carried out. This is attestation, not verification-there is still no automatic confirmation mechanism anywhere in the system-and the ESG report that aggregates these records is explicitly a decision log, not a verified-savings report.

3.11 ROLE-BASED ACCESS CONTROL

Three roles are enforced both server-side, via a dependency checked on every mutating endpoint, and client-side, via route guards and conditional navigation.

Table 3.8 Role-Based Access Control Matrix

Role	Can do
infrastructure_manager	Everything-accept/snooze/dismiss recommendations, edit preferences, manage users
sustainability_manager	Read-only-dashboards and the ESG report
operations_engineer	Mark approved actions as executed; cannot make the original accept/snooze/dismiss decision

Authentication is JWT-based, with users seeded through a dedicated script.

3.12 DATABASE DESIGN

PostgreSQL is used in deployment, with SQLite available as a zero-code-change default controlled by a single environment variable. Schema changes are managed through Alembic, baselined onto the pre-existing schema. Eleven tables make up the schema: **users, servers (static metadata), four raw telemetry tables (server_telemetry, power_telemetry, cooling_telemetry, storage_telemetry), forecasts (a persisted forecast audit log), flags, recommendations, operator_actions, and preferences.**

CHAPTER 4

IMPLEMENTATION

4.1 TECHNOLOGY STACK

- **Backend:** Python, FastAPI, Pydantic (validation), SQLAlchemy (ORM), Alembic (migrations).
- **Database:** PostgreSQL in deployment, SQLite for local development, selected via the DATABASE_URL environment variable.
- **Machine learning:** scikit-learn (Linear Regression, Random Forest, preprocessing pipeline), XGBoost.
- **Frontend:** React, Vite, react-router-dom, recharts.
- **Authentication:** JSON Web Tokens (JWT).
- **Simulation:** independent Python scripts per telemetry domain, run as long-lived processes..

4.2 SYSTEM REQUIREMENTS

4.2.1 HARDWARE REQUIREMENTS

Table 4.1 Hardware Requirements

Component	Minimum Recommendation	Purpose
Processor	Dual-core 2 GHz or better	Runs the backend, frontend, and simulators.
Memory	8 GB RAM	Supports PostgreSQL, Python packages, browser, and model training.
Storage	At least 5 GB free space	Stores source code, dependencies, database, logs, and model artifacts.

Network	Localhost network access	Allows frontend, backend, and simulator processes to communicate.
Display	1280 × 720 or better	Recommended for dashboard and analytics review.

4.2.2 SOFTWARE REQUIREMENTS

Table 4.2 Software Requirements

Software	Recommended role
Python 3.10 or later	Backend, simulators, training, and tests.
Node.js and npm	React/Vite frontend development.
PostgreSQL 14 or later	Production-style relational database.
Git	Source-code version control.
Visual Studio Code	Local development environment.
Modern web browser	Dashboard access and UI testing.

The project is installed locally using the following commands:

```
None
cd greenops
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r backend/requirements.txt
cd frontend
npm install
```

On Windows PowerShell, the Python environment is activated using:

```
None
.venv\Scripts\Activate.ps1
```

4.3 DATABASE, REACT FRONTEND, TELEMETRY, AND API INTEGRATION

DATABASE CONFIGURATION

PostgreSQL is created using the following SQL commands:

SQL

```
CREATE DATABASE greenops; CREATE USER greenops_user WITH PASSWORD 'change_this_password';  
GRANT ALL PRIVILEGES ON DATABASE greenops TO greenops_user;
```

The backend reads the database connection from an environment variable.

SQL

```
DATABASE_URL="postgresql+psycpg2://greenops_user:change_this_password@localhost:5432/greenops"
```

The main database tables store users, server metadata, server telemetry, power telemetry, cooling telemetry, storage telemetry, forecasts, flags, recommendations, operator actions, and priority preferences. Raw telemetry and decision records are stored in the database, while serialized machine-learning model files are stored as artifacts rather than as binary database columns.

SQLAlchemy maps Python classes to database tables. A telemetry record is first validated by Pydantic, then converted into a SQLAlchemy object, added to the active session, committed, and returned as a JSON response. Alembic is used to manage controlled schema changes when the database structure evolves.

REACT AND VITE FRONTEND

The React interface is built with Vite. Vite provides the development server, module bundling, fast refresh, and production build process. In the existing project, the frontend is already initialized; its dependencies are installed with:

None

```
cd frontend  
npm install
```

```
//The frontend development server is started using:  
npm run dev
```

The frontend displays values supplied by the backend rather than independently recalculating authoritative metrics. This keeps the backend as the source of truth and avoids inconsistent values between screens. Recharts is used to display time-series trends, server comparisons, utilization changes, energy patterns, and forecast results.

TELEMETRY SIMULATOR EXECUTION

Four independent Python processes generate the telemetry used during the demonstration. They are started in separate terminals after the backend is available.

```
None  
python simulators/server_monitor.py  
python simulators/power_monitor.py  
python simulators/cooling_monitor.py  
python simulators/storage_monitor.py
```

API INTEGRATION

The API is divided into functional groups. Telemetry endpoints receive records from the simulators. Server endpoints return current state and historical records. Analytics endpoints calculate time-window summaries. Forecast endpoints return future workload estimates. Recommendation endpoints return ranked actions and what-if details. Operator-action endpoints store human decisions. Report endpoints return ESG and decision-log information.

Table 4.3 Primary REST Endpoint Reference

Endpoint	Purpose
POST /telemetry/server	Ingest server telemetry
POST /telemetry/power	Ingest power telemetry
POST /telemetry/cooling	Ingest cooling telemetry
POST /telemetry/storage	Ingest storage telemetry
POST /servers/register	Register a new server before it can post telemetry
GET /servers	Current state of all servers
GET /servers/{id}/detail	Combined live view for one server, used by the Server Detail page
GET /servers/{id}/history	Historical telemetry for one server
GET /servers/{id}/prediction	ML CPU and power estimate vs measured
GET /servers/{id}/forecast	One-hour, six-hour or twenty-four-hour CPU projection
GET /forecasts	Persisted forecast history and validation metrics
GET /flags	All active rule-based flags
GET /storage/flags	Storage-specific flags only
GET /thresholds	Per-server-type idle thresholds and severity weights
GET /analytics/summary	PUE, WUE, energy, cost and carbon overview
GET /analytics/daily	Daily time-series data behind the Analytics page's charts
GET /analytics/history	Longer-range historical analytics data
GET /recommendations	Ranked pending recommendations
GET /recommendations/{id}	Full detail for one recommendation, without re-running the simulation
GET /recommendations/{id}/what-if	Consolidation safety and impact simulation
POST /recommendations/{id}/action	Record the operator's accept, snooze or dismiss decision
GET /operator-actions	Decision history for the Approved Actions page
PATCH /operator-actions/{id}	Attest that an approved action was executed
GET /preferences	Read the current recommendation-ranking weights

Endpoint	Purpose
PUT /preferences	Update the recommendation-ranking weights
GET /reports/esg	Aggregated ESG summary over a date range
GET /reports/esg/export	ESG report as CSV
GET /reports/esg/export/pdf	ESG report as PDF
GET /model/evaluation	Three-model comparison metrics for CPU and power estimation
POST /auth/login	Authenticate and receive a session token
GET /auth/me	Return the currently authenticated user's role and details
GET /users, POST /users, PATCH /users/{id}, DELETE /users/{id}	Manage user accounts and roles, infrastructure manager only

4.4 MACHINE-LEARNING, FORECASTING, RECOMMENDATION, AND DECISION IMPLEMENTATION

The machine-learning layer contains two related but separate tasks. The CPU estimation model estimates the current CPU utilization from current workload and environmental features. The workload-forecasting model predicts a future CPU or workload value using timestamped lag features. The CPU model is not silently renamed as the forecasting model; the two tasks have different targets and input structures.

The offline training process loads the historical telemetry dataset, handles missing values, encodes categorical features, scales numeric features where required, constructs the training matrix, and splits the data chronologically. Linear Regression, Random Forest, and XGBoost are compared using MAE, RMSE, and R^2 . The selected model and its evaluation metrics are saved for later inference.

A forecast generated at 10:00 for 10:15 is not considered validated immediately. When telemetry for 10:15 becomes available, the predicted value is compared

with the actual value. The system can then calculate the absolute error, squared error, and aggregate MAE or RMSE over many forecasts.

The recommendation engine begins with active flags. An idle-server flag may produce a consolidation recommendation, stale data may produce an archive recommendation, duplicate data may produce a deduplication recommendation, and over-provisioned storage may produce a rightsizing recommendation. For consolidation, the what-if engine evaluates candidate target servers by adding the source workload with an overhead factor. It checks CPU, memory, network, and thermal safety limits in both the current state and the near-term forecast state.

The recommendation score combines normalized energy, cost, carbon, water, and risk impacts using configurable weights. The recommended action is not automatically executed. The human decision interface allows an authorized operator to approve, snooze, dismiss, or reject the recommendation. The system stores the selected decision, user identity, timestamp, score, candidate details, and impact snapshot in an audit record.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 RULE ENGINE RESULTS

With the thirteen simulated servers running continuously, the rule engine correctly identifies the servers deliberately biased toward idle (S4, S5, S6, S8, S10, S12 in Table 3.1) within the first few minutes of telemetry collection and correctly assigns the higher severity weight to idle GPU servers (S8) relative to idle servers of other types at a comparable CPU percentage, matching the per-type thresholds derived in Section 3.4. Flags resolve automatically once a server's rolling average CPU rises back above its type's threshold and no duplicate recommendation accumulates when a server flaps between idle and healthy across consecutive evaluation windows, confirming the intended behaviour of the flag-synchronization routine.

5.2 ML MODEL EVALUATION RESULTS

Both estimation models were trained on the real, 10,000-row “Green AI Data Center Telemetry” dataset (15 columns, with missing values only in `datacenter_region`, imputed with the most frequent category inside the preprocessing pipeline). Tables 5.1 and 5.2 report every candidate algorithm's result. In both cases Linear Regression was selected since neither Random Forest nor XGBoost cleared the pre-registered 5% relative MAE improvement bar over it. Figure 5.1 compares the MAE values across the evaluated models and provides a visual comparison of the model performance.

Table 5.1 CPU Estimation Model: Candidate Comparison
(Target: cpu_utilization)

Model	MAE	RMSE	R ²
Linear Regression (selected)	4.511	7.671	0.845
Random Forest	4.828	8.077	0.828
XGBoost	4.833	7.999	0.831

Table 5.2 Power Estimation Model: Candidate Comparison
(Target: power, rescaled to 1.2–4.5 kW)

Model	MAE	RMSE	R ²
Linear Regression (selected)	0.211	0.330	0.422
Random Forest	0.227	0.346	0.366
XGBoost	0.222	0.341	0.385

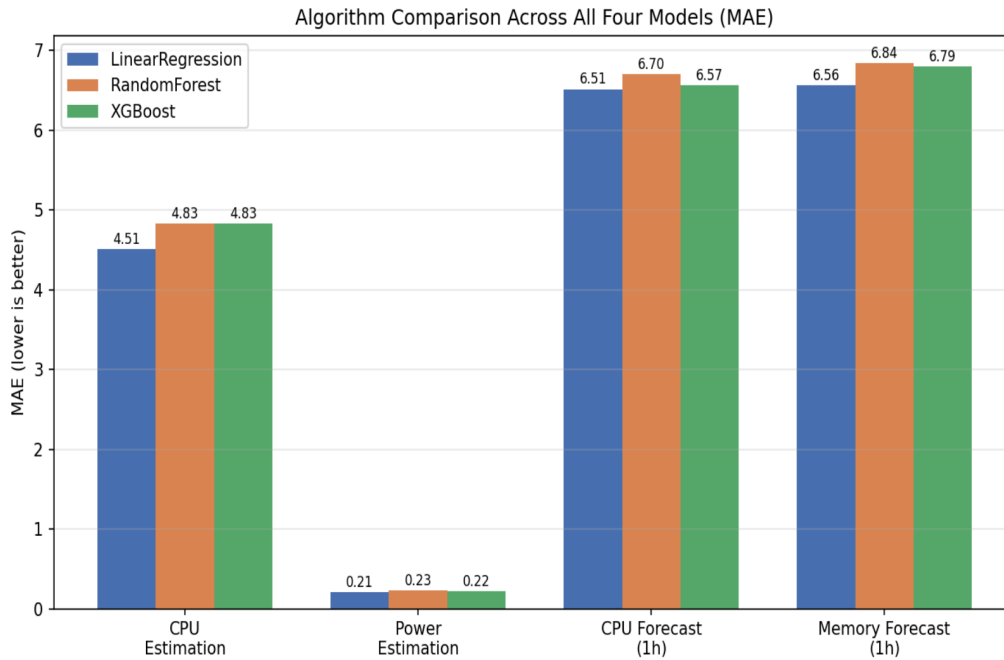


Fig 5.1. Comparison of evaluated algorithms across CPU estimation, power estimation, CPU forecasting and memory forecasting using MAE

The CPU model explains the large majority of variance in the target ($R^2 = 0.845$) using only workload and environmental features, with its own target correctly excluded to avoid leakage. The power model is noticeably weaker ($R^2 = 0.422$ at best) even with `cpu_utilization` included as an input. This is a genuine and expected finding rather than a bug and indicates that power draw in the real dataset depends on hardware and efficiency factors not fully captured by the available feature set. This asymmetry between the two targets is consistent with the general pattern reported in the wider literature on data-center power modelling (Section 2.4) and directly informed the decision to make the power model's prediction the primary basis for the what-if energy estimate, with a simpler CPU-ratio formula used only as a fallback when the model or the target's cooling telemetry is unavailable. Every number produced this way is explicitly labelled as an estimate rather than presented as a measurement. Figure 5.2 further illustrates this performance through the predicted-versus-actual plot, where the predictions show a close correspondence with the actual CPU utilization values across the held-out test set.

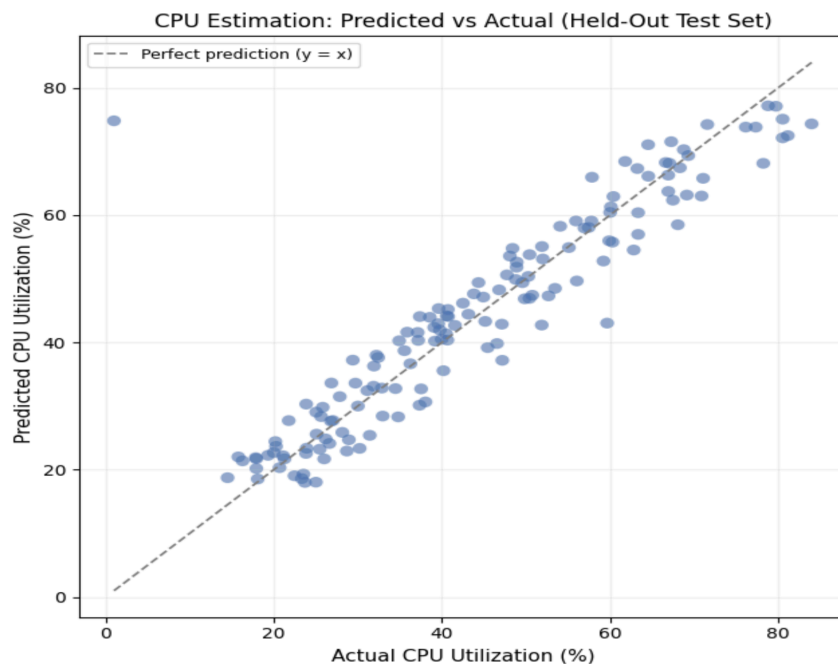


Fig. 5.2. CPU Estimation: Predicted vs Actual

5.3 WHAT-IF SIMULATION FOR SERVER CONSOLIDATION

Consider server S6 (Compute, workload bias 0.05, “very idle”), which is reliably flagged `idle_server` once its six-hour rolling average CPU falls below the Compute-type threshold of 17.3%. Requesting a what-if simulation for S6 causes the engine to enumerate every other Compute-type server not itself idle-flagged. In this fleet that means S1, S2 and S13. The engine computes each one's post-move CPU, memory and network as its own current load plus 90% of S6's current load. Each candidate is then checked against the four safety dimensions in Table 3.7. A candidate already running close to its own capacity or whose predicted post-move inlet temperature would exceed 30°C is excluded from `safe_now` regardless of headroom on other dimensions. Among the remaining candidates, the one with the most spare headroom after the move is chosen as the recommended target and its own fifteen-minute-ahead CPU and memory forecast is checked in exactly the same way before the recommendation is finalized as safe. The resulting recommendation card shows the chosen target, the full list of considered candidates with their pass/fail status on each dimension and an estimated facility-power impact derived from the trained power model, with the estimate's assumptions (which model was used and that it is a legitimate estimate rather than a measurement) stated explicitly alongside the number, consistent with the honesty principle followed throughout the system.

5.4 HONEST SUMMARY OF SYSTEM MATURITY

Consistent with the principle of never presenting an estimate with the same confidence as a measurement, the current system can be summarized in three honest categories.

5.4.1 GENUINELY REAL

The rule engine (pure threshold logic, no ML involved), PUE (a direct ratio of two independently simulated values), the offline ML validation results reported in Tables 5.1 and 5.2, the timestamp-based energy integration, and the four-dimension consolidation safety check.

5.4.2 LEGITIMATE ESTIMATES, CLEARLY LABELLED

Cost, carbon, water, and the power model's what-if energy prediction, which agrees with the simulator on direction and scale but will not match it exactly, since the two are independently built approximations of the same underlying quantity.

5.4.3 DELIBERATELY NOT BUILT

Criticality and SLA-aware gating, target storage-capacity and electrical/PDU-capacity safety checks, a verified (rather than operator-attested) confirmation mechanism for executed actions, the CPU model's designed hardware-health-check role (which needs environmental-fault injection in the simulator first), and trained six-hour/twenty-four-hour forecast models, for which there is not yet enough real telemetry history.

5.5 SYSTEM IMPLEMENTATION

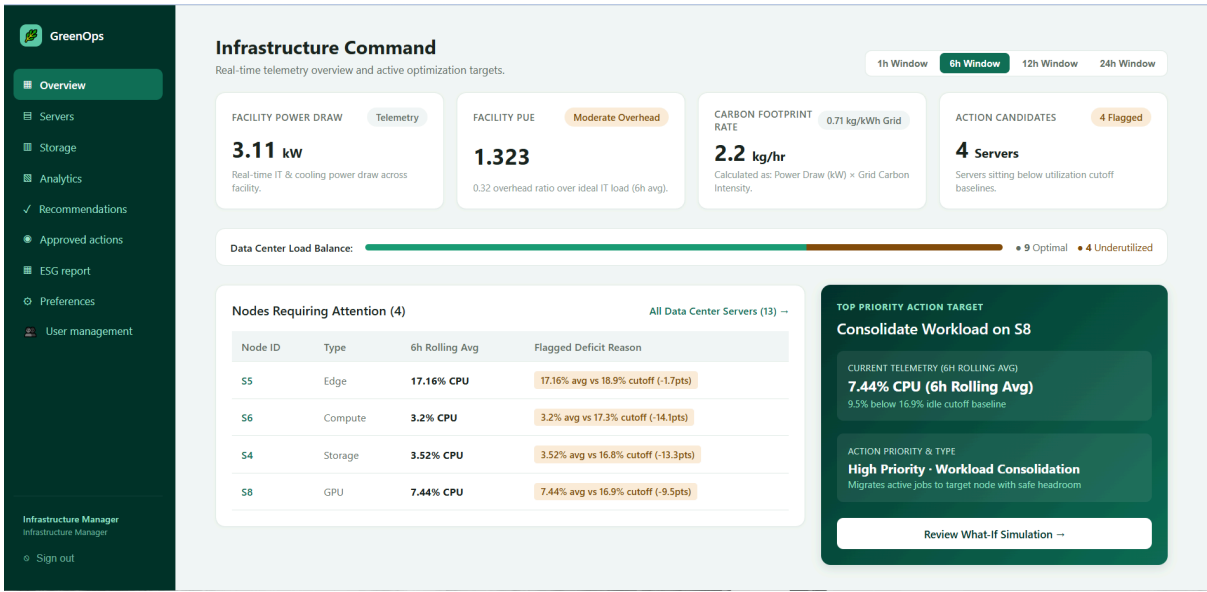


Fig. 5.3. Dashboard Overview Page

The Overview page summarizes facility wide values for PUE, cumulative energy, carbon and the current count of flagged servers being refreshed from live simulated telemetry.

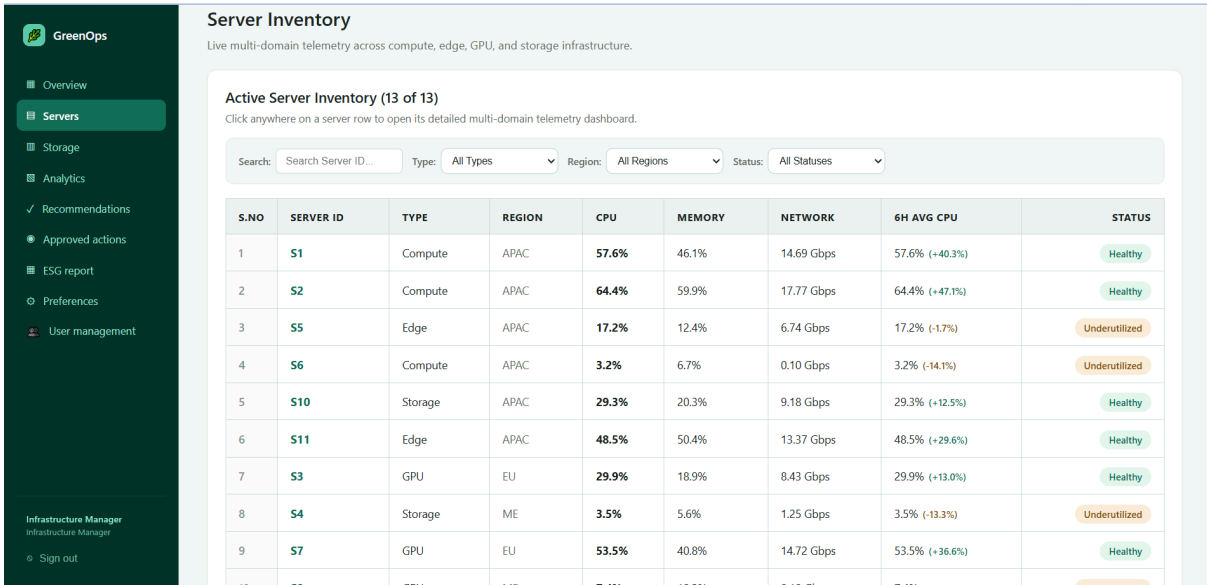


Fig. 5.4. Servers Details Page

The Server detail page gives insights into per-server view combining live environmental reading, the CPU forecast and any active waste flags for that specified server.

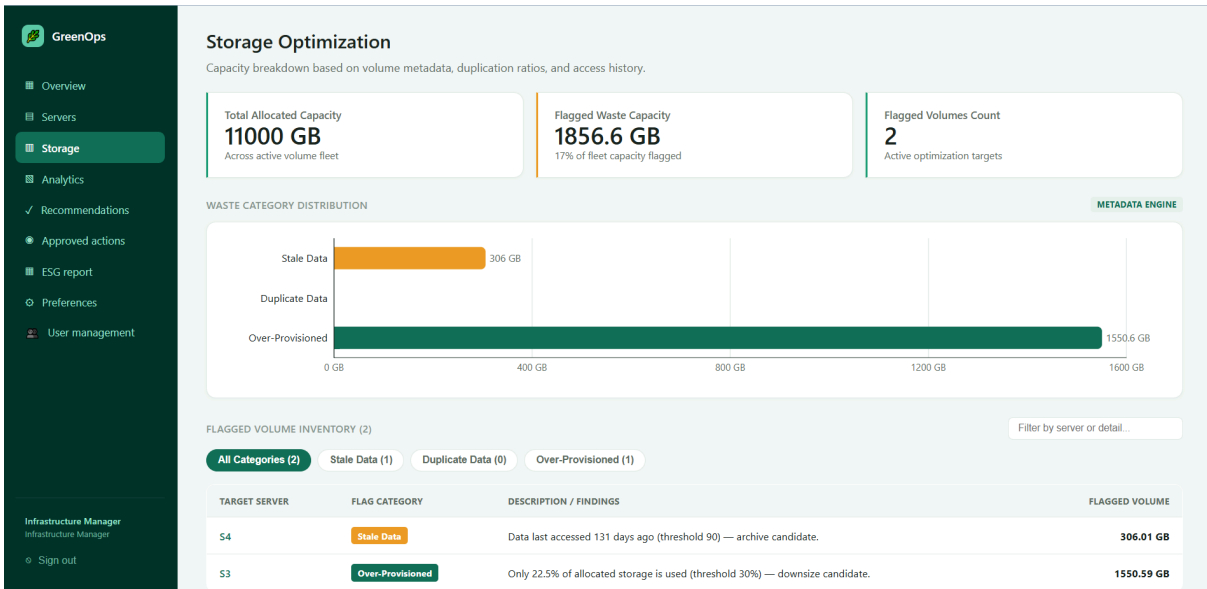


Fig. 5.5. Storage Details Page

The Storage page shows specific waste view, surfacing stale, duplicated and over-provisioned volumes flagged by the rule engine.

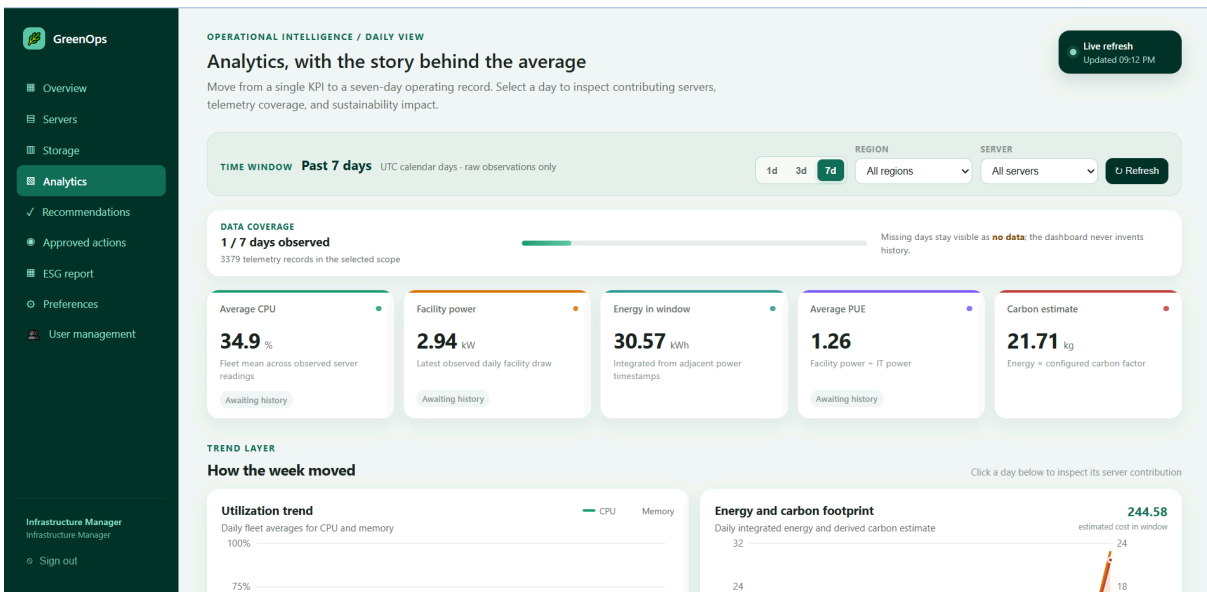


Fig. 5.6. Analytics Page

The Analytics page breaks down the facility-level of idle servers, power/cooling performance and storage waste over time which is used to spot trends.

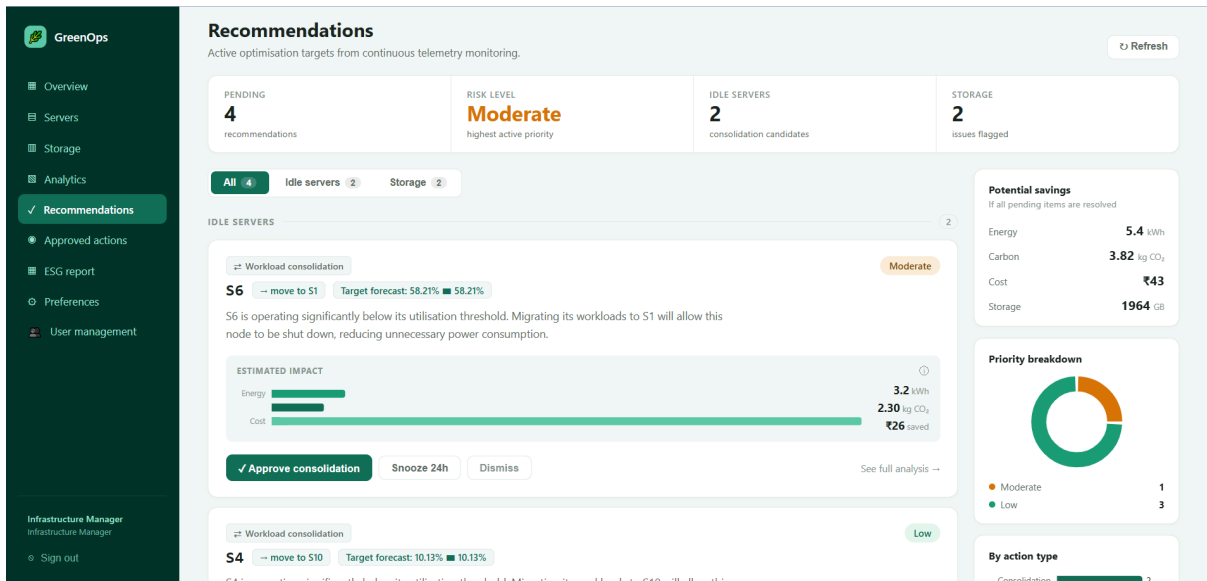


Fig. 5.7. Recommendations Page

The Recommendations page ranks a list of pending consolidate/rightsize actions with inline Accept / Snooze / Dismiss controls, mini impact charts and a sidebar summarizing pending savings across the batch.

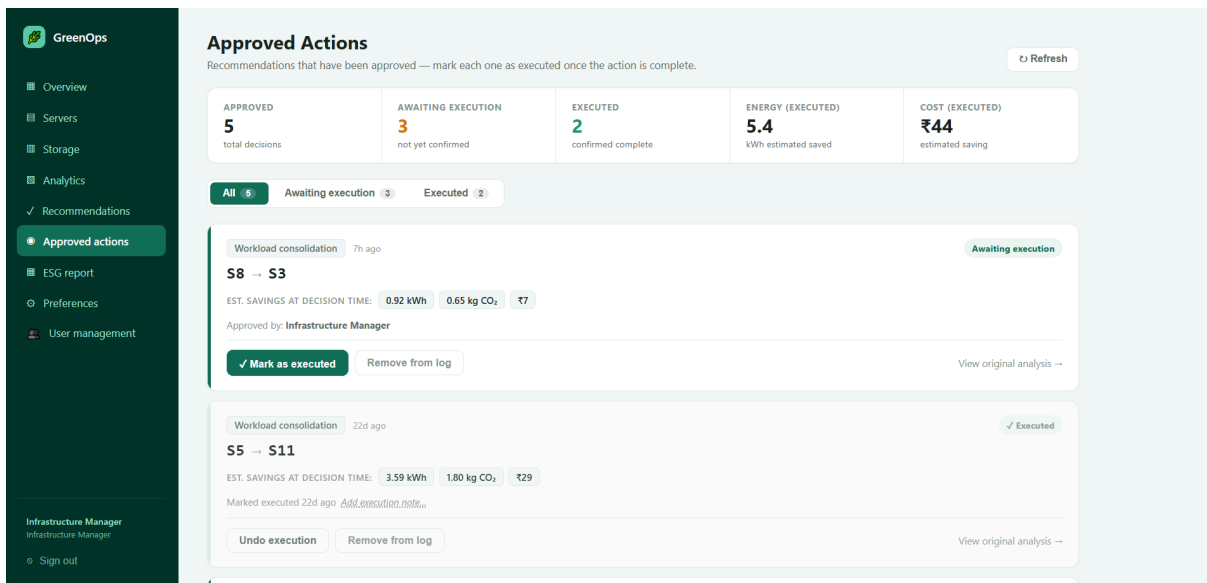


Fig. 5.8. Approved Actions Page

The Approved actions logs the past operators decisions, filterable by executed vs awaiting executions where an operator can retroactively attach an executed at timestamp and a note once the action is carried out.

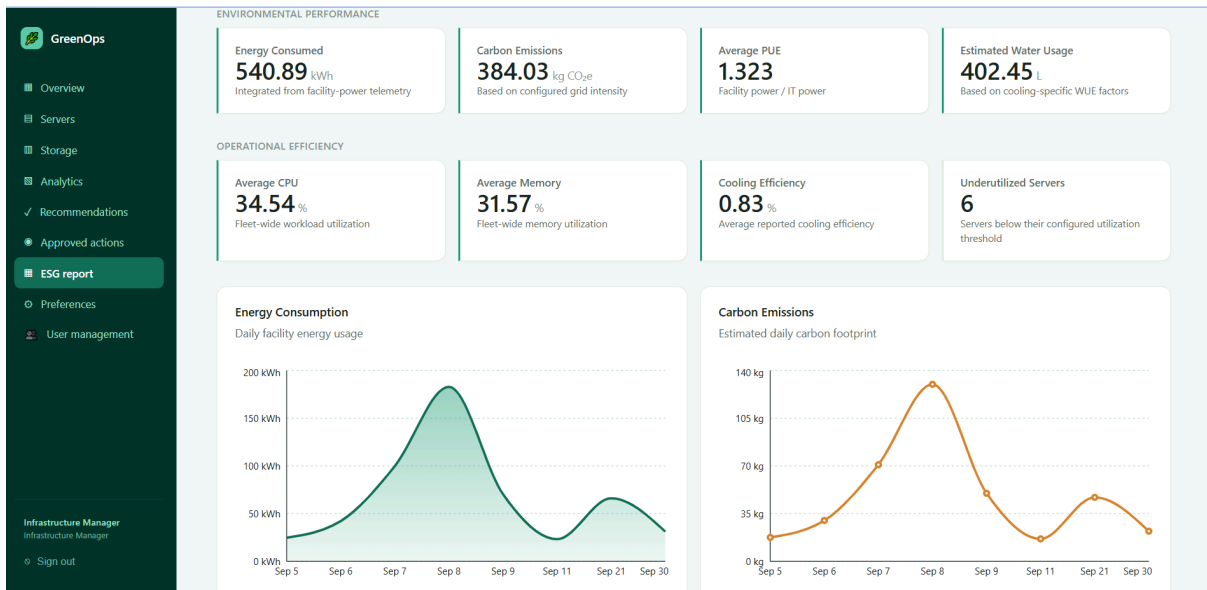


Fig. 5.9. ESG Reporting Page

This page has aggregated log of operator-action over a selected date range, exportable as CSV or PDF, presented explicitly as a decision record.

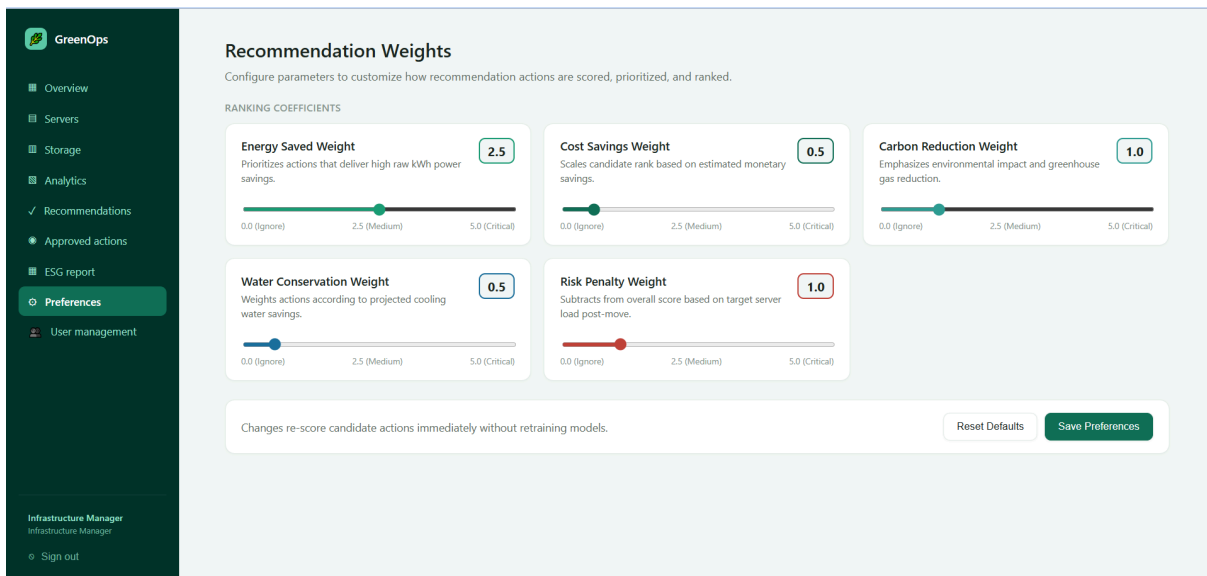


Fig. 5.10. Preferences Page

The Preferences page has a weight slider for cost, carbon, water and risk that feeds directly into the recommendation scoring formula, re-ranking pending recommendations instantly without retraining any model.

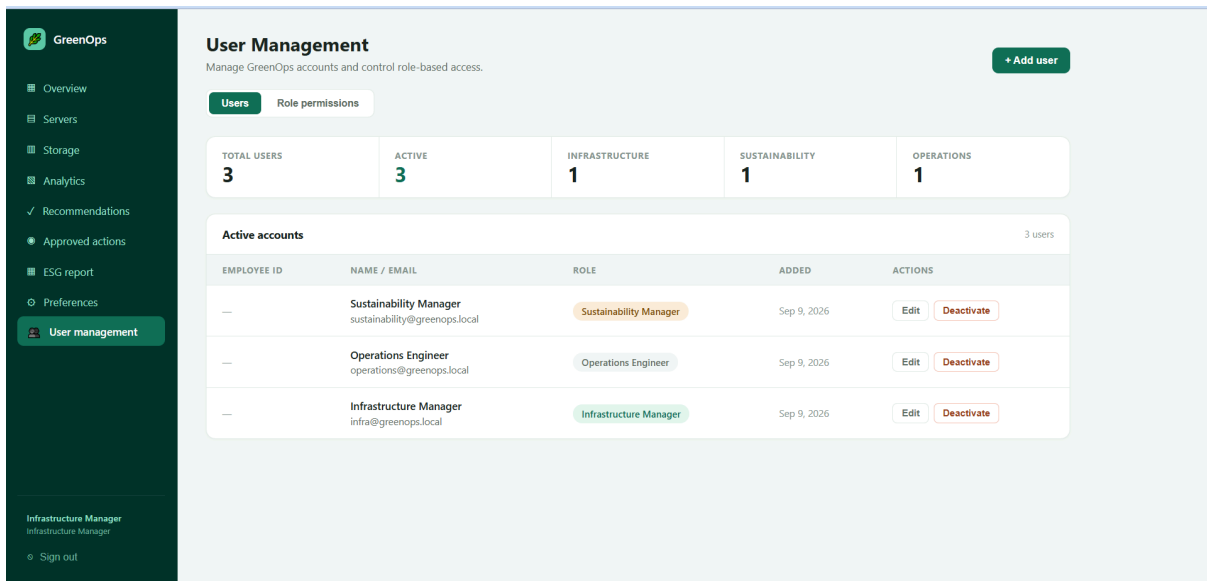


Fig. 5.11. User Management Page

The User Management page provides administrative controls for managing GreenOps accounts and role-based access.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this project, we have discussed the problem of inefficient resource utilization and sustainability in on-premise data centers and have developed GreenOps, a decision-support platform that would help in monitoring infrastructure resources and identifying possible optimization opportunities. The proposed system combines telemetry monitoring, sustainability calculation, workload forecasting and recommendation generation to provide a unified solution for managing underutilized servers in on-premise environments. It has been noted that the functioning of the system involves multiple stages, starting from telemetry collection and identification of underutilized resources, followed by workload prediction and what-if analysis before providing a recommendation to the operator. The system calculates important sustainability measures such as energy, carbon, water and cost, along with PUE and WUE, and uses statistically derived thresholds to identify inefficient resource usage. The implemented models achieved an R^2 value of 0.845 for CPU estimation and 0.422 for power estimation. The recommendation engine further allows different priorities to be assigned to sustainability and performance factors before suggesting suitable actions. The proposed system does not automatically execute any infrastructure changes, and the final decision remains with the human operator. Thus, GreenOps provides a transparent and priority-aware approach for supporting resource optimization and sustainability management in on-premise data center environments.

6.2 LIMITATIONS

The proposed system provides a working platform for resource monitoring and optimization in on-premise data centers, but its real-world implementation poses certain challenges. The telemetry used in the current system is simulated rather than collected from actual data-center hardware, and the cooling simulation does not currently represent independent equipment failures. The cost, carbon and water calculations are also based on configured values rather than live external data. The what-if analysis does not yet consider certain factors such as storage capacity, electrical capacity, workload criticality and SLA requirements. In addition, longer-term forecasting for twenty-four-hour periods has not been implemented due to the limited historical data available, and approved actions are currently confirmed by the operator rather than independently verified. Storage optimization and ESG reporting also remain as areas for further development.

6.3 FUTURE WORK

The proposed system can be further enhanced by replacing the simulated telemetry with real-time data from on-premise data-center infrastructure through sources such as Prometheus, hypervisor APIs, smart PDU and BMS systems. The cooling component can also be extended to support controlled environmental fault conditions, while live electricity tariffs, regional carbon-intensity information and actual water measurements can be incorporated to improve the sustainability calculations. The what-if analysis can be further developed by considering storage capacity, electrical capacity, workload criticality and SLA requirements before generating recommendations. With the availability of sufficient historical telemetry, a dedicated 24-hour workload forecasting model can also be developed to support longer-term resource optimization decisions. An independent verification mechanism can be

introduced to confirm approved actions and measure the actual savings achieved. The existing storage optimization and ESG reporting capabilities can also be further enhanced with additional analytics and reporting features, allowing GreenOps to provide a more comprehensive sustainability and resource-management solution for on-premise data centers.

APPENDIX 1 - KEY GENERATING FORMULAS

The following formulas, reproduced from the simulator implementation, generate every live telemetry reading in the system.

A1.1 SERVER_MONITOR.PY

$$workload = clamp(0.02, 0.98, gauss(bias, 0.08))$$

$$cpu = clamp(1, 99, workload \times 100 + uniform(-5, 5))$$

$$memory = clamp(5, 99, cpu \times 0.9 + uniform(-8, 8))$$

$$network = max(0.1, workload \times 25 + uniform(-2, 2))$$

A1.2 POWER_MONITOR.PY

$$variable_kw = 3.3 \times (0.85 \times cpu_fraction + 0.15 \times memory_fraction)$$

$$it_power_kw = max(0.3, 1.2 + variable_kw + uniform(-0.15, 0.15))$$

$$penalty_fraction = (0.99 - cooling_efficiency) \div (0.99 - 0.4)$$

$$overhead_ratio = max(1.05, 1.15 + 0.40 \times penalty_fraction + uniform(-0.03, 0.03))$$

$$facility_power_kw = it_power_kw \times overhead_ratio$$

A1.3 COOLING_MONITOR.PY

$$inlet_temperature_c = 19.0 + temp_rise_by_type[type] \times cpu_fraction + uniform(-0.8, 0.8)$$

$$cooling_efficiency = clamp(0.4, 0.99, 0.95 - efficiency_drop_by_type[type] \times cpu_fraction + uniform(-0.03, 0.03))$$

where $temp_rise_by_type = \{Air: 9.0, Hybrid: 5.0, Liquid: 2.5\}$ °C at 100% CPU, and $efficiency_drop_by_type = \{Air: 0.35, Hybrid: 0.20, Liquid: 0.10\}$.

A1.4 CONSOLIDATION POST-MOVE LOAD

$$post_move_cpu = target.current_cpu + source.current_cpu \times 0.9$$

$$post_move_memory = target.current_memory + source.current_memory \times 0.9$$

$$post_move_network = target.current_network + source.current_network \times 0.9$$

$$post_move_temp_c = 19.0 + temp_rise_by_type[target.type] \times \min(1.0, post_move_cpu / 100)$$

A1.5 RECOMMENDATION RANKING SCORE

$$score = \sum (weight_i \times normalized_impact_i) - risk_weight \times risk$$

APPENDIX 2 - DATABASE SCHEMA

The following eleven tables make up the shared PostgreSQL database described in Section 3. Nullable columns are marked accordingly. Timestamp columns default to the moment of insertion unless otherwise noted.

A2.1 USERS

Column	Type	Notes
id	Integer	Primary key
employee_id	String	Unique, nullable
name	String	Not null
email	String	Unique, not null
password_hash	String	infrastructure_manager, sustainability_manager or operations_engineer
is_active	Boolean	Default true
created_at	DateTime	Default current time

A2.2 SERVERS

Column	Type	Notes
id	Integer	Primary key
server_id	String	Unique, e.g. S1 through S13
server_type	String	Default Compute
cooling_type	String	Default Air
datacenter_region	String	Default APAC

A2.3 SERVER_TELEMETRY

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
cpu_utilization	Float	
memory_utilization	Float	
network_throughput_gbps	Float	
workload_intensity	Float	
timestamp	DateTime	Indexed

A2.4 POWER_TELEMETRY

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
it_power_kw	Float	
facility_power_kw	Float	
timestamp	DateTime	Indexed

A2.5 COOLING_TELEMETRY

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
inlet_temperature_c	Float	
cooling_efficiency	Float	
cooling_type	String	
timestamp	DateTime	Indexed

A2.6 STORAGE_TELEMETRY

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
total_storage_gb	Float	
used_storage_gb	Float	
duplicate_data_gb	Float	
last_accessed_days_ago	Integer	
storage_type	String	
timestamp	DateTime	Indexed

A2.7 FORECASTS

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
horizon_minutes	Integer	60,360 or 1440 for the dashboard tier
measured_cpu	Float	
predicted_cpu	Float	Nullable when there is not enough history
method	String	e.g. LinearRegression or historical_average
model_version	String	Nullable
validation_mae	Float	Nullable
validation_rmse	Float	Nullable
validation_r2	Float	Nullable
source_observations	Integer	Default 0
generated_at	DateTime	Indexed

A2.8 FLAGS

Column	Type	Notes
id	Integer	Primary key
server_id	String	Indexed
flag_type	String	idle_server, stale_data, duplicate_data or overprovisioned
severity	String	low, medium or high
detail	String	Human-readable explanation
metric_value	Float	e.g. average CPU or GB reclaimable
created_at	DateTime	Indexed
resolved	Integer	0 active, 1 resolved

A2.9 RECOMMENDATIONS

Column	Type	Notes
id	Integer	Primary key
flag_id	Integer	Indexed, unique, not null
server_id	String	Indexed, not null
recommendation_type	String	consolidate, archive, deduplicate or rightsize
priority	String	high, medium or low
explanation	String	Not null
status	String	pending, accepted, rejected or snoozed
created_at	DateTime	Indexed
decided_at	DateTime	Nullable
snoozed_until	DateTime	Nullable

A2.10 OPERATOR_ACTIONS

Column	Type	Notes
id	Integer	Primary key
recommendation_id	Integer	Indexed, not null
server_id	String	Indexed, not null

Column	Type	Notes
action	String	consolidate, rightsize, snooze or do_nothing
notes	String	Nullable
estimated_energy_saving_kwh	Float	Nullable, snapshot at decision time
estimated_carbon_reduction_kg	Float	Nullable, snapshot at decision time
estimated_cost_saving	Float	Nullable, snapshot at decision time
estimated_storage_reclaimed_gb	Float	Nullable, snapshot at decision time
target_server_id	String	Nullable, the chosen consolidation target if any
executed_at	DateTime	Nullable, null means approved but not yet executed
execution_note	String	Nullable
decided_by_user_id	Integer	Foreign key to users.id, nullable
decided_by_name	String	Nullable, snapshot at decision time
decided_by_role	String	Nullable, snapshot at decision time
created_at	DateTime	Indexed

A2.11 PREFERENCES

Column	Type	Notes
id	Integer	Primary key, single row by convention
energy_weight	Float	Default 1.0
cost_weight	Float	Default 1.0
carbon_weight	Float	Default 1.0
water_weight	Float	Default 0.5
risk_weight	Float	Default 1.0
updated_at	DateTime	Updated on every write

REFERENCES

- [1] A. Ahmed, D. Brown, and A. Gegov, “Dynamic resource allocation through workload prediction for energy efficient computing,” in *Advances in Computational Intelligence Systems*, 2017, pp. 35–44.
- [2] M. Alex, S. O. Ojo, and F. M. Awuor, “Carbon-aware, energy-efficient, and SLA-compliant virtual machine placement in cloud data centers using deep Q-networks and agglomerative clustering,” *Computers*, vol. 14, no. 7, Art. no. 280, 2025.
- [3] J. Bawa, K. K. Chahal, and K. Kaur, “Improving cloud resource management: An ensemble learning approach for workload prediction,” *The Journal of Supercomputing*, vol. 81, no. 10, Art. no. 1138, 2025.
- [4] A. Beloglazov, J. Abawajy, and R. Buyya, “Energy-aware resource allocation heuristics for efficient management of data centers for cloud computing,” *Future Generation Computer Systems*, vol. 28, no. 5, pp. 755–768, 2012.
- [5] European Commission, “Corporate sustainability reporting,” *Financial Markets, Company Reporting and Auditing*.
- [6] J. Gorey, “Data drain: The land and water impacts of the AI boom,” *Land Lines Magazine*, Lincoln Institute of Land Policy, Oct. 17, 2025.
- [7] T. Khan, W. Tian, S. Ilager, and R. Buyya, “Workload forecasting and energy state estimation in cloud data centres: ML-centric approach,” *Future Generation Computer Systems*, vol. 128, pp. 320–332, 2022.

- [8] G. Lovász, F. Niedermeier, and H. de Meer, “Performance tradeoffs of energy-aware virtual machine consolidation,” *Cluster Computing*, vol. 16, no. 3, pp. 481–496, 2013.
- [9] M. Grochulska-Salak et al., “Green data centres: Sustainable solutions with green energy and green–blue infrastructure,” *Energies*, vol. 18, no. 24, Art. no. 6592, 2025.
- [10] A. Radovanovic et al., “Carbon-aware computing for datacenters,” *arXiv:2106.11750*, 2021.
- [11] Recharts, “Recharts: A composable charting library built on React components,” Recharts Documentation.
- [12] React, “React: The library for web and native user interfaces,” React Documentation.
- [13] Scaphandre, “Scaphandre: A metrology agent dedicated to electrical power consumption,” Scaphandre Documentation.
- [14] C. Subramanian, A. Vasan, and A. Sivasubramaniam, “Reducing data center power with server consolidation: Approximation and evaluation,” in *Proc. 17th Int. Conf. High Performance Computing (HiPC)*, 2010.
- [15] Y. Tao and P. Gao, “Global data center expansion and human health: A call for empirical research,” *Eco-Environment & Health*, vol. 4, no. 3, Art. no. 100157, 2025.
- [16] V. Tripathi et al., “Scaling India’s data centre ecosystem: Stakeholder perspectives on infrastructure, energy, and resilience,” *Council on Energy, Environment and Water and SYSTEMIQ*, 2026.

- [17] M. Uddin et al., “Server consolidation: A technique to enhance cloud data center power efficiency and overall cost of ownership,” *International Journal of Distributed Sensor Networks*, vol. 17, no. 3, 2021.
- [18] V. Vaccaro, L. C. Tagliabue, and M. Aldinucci, “Sustainable data centers: Advancing energy efficiency and resource optimization,” in *Proc. 33rd Euromicro Int. Conf. Parallel, Distributed and Network-Based Processing (PDP)*, Turin, Italy, 2025, pp. 486–493.
- [19] Vite, “Vite: Next Generation Frontend Tooling,” Vite Documentation.
- [20] “Green AI Data Center Telemetry” dataset, Kaggle.
- [21] X. Zhan et al., “Data center cooling system optimization using offline reinforcement learning,” *arXiv:2501.15085*, 2025.
- [22] M. R. R. Deva and N. K. Jain, “Utilizing Azure Automated Machine Learning and XGBoost for predicting cloud resource utilization in enterprise environments,” in *Proc. 2025 Int. Conf. Networks and Cryptology (NETCRYPT)*, 2025, pp. 535–540.
- [23] OpenCost, “OpenCost: Open source cost monitoring for cloud native environments,” OpenCost Documentation.

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Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization

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Abstract— On-premises data centers today operate around the clock to guarantee availability but a large share of the electricity, cooling capacity and water they have to consume goes toward servers that are barely being used. This stays hidden because the telemetry that could reveal it are typically scattered across disconnected monitoring tools that were never built to talk to each other. Our work addresses this gap through GreenOps, an observability and resource intelligence platform that unifies this data in one integrated pipeline and makes it intelligible to human operators. The system collects live telemetry (simulated in this prototype) from server, power, cooling and storage monitors and stores it with full historical context and computes real time sustainability metrics including Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), energy cost and carbon emissions. GreenOps employs four machine learning models - CPU estimation, power estimation, workload forecasting and memory workload forecasting each trained from related telemetry. For every model, three candidate algorithms (Linear Regression, Random Forest and XGBoost) are evaluated honestly on held out data giving operators both a real-time and a near-future predictive view. Idle servers, stale storage and over-provisioned resources get flagged by a rule-based detection layer. If a server is flagged idle, a forecasting layer projects its near term and a what-if simulation engine trials possible migrations and estimates the savings involved. A priority-weighted recommendation engine then ranks these against cost, carbon and performance risk. A React based dashboard surfaces all of this, built for a human to review rather than for automation to act on unsupervised. This paper describes the system's architecture, the reasoning behind the modeling choices and results from our working prototype.

Keywords— *GreenOps, On-premises Data Centers, Sustainability, Machine Learning, Resource Optimization, Workload Forecasting, PUE, WUE*

I. INTRODUCTION

Data Centers have quietly become one of the most power-hungry categories of infrastructure in the modern world. Whether a website is being loaded, a model is being trained, or a file is being synced to the cloud, it all traces back to a room full of servers running continuously throughout the day, regardless of whether anyone is actually using them at that moment. This 24/7 operation is by design, as availability is the goal, but it occurs at a steep environmental cost. The facilities that support this architecture also carry enormous overhead in the form of cooling systems, backup power infrastructure and

water usage all of which is driven by how much hardware is kept running rather than by how useful the work it performs is.

A significant portion of this consumption is completely avoidable. Servers frequently sit at low utilization for long periods of time, often because they were sized for a peak that rarely arrives, or because the workload they were built for moved on and nobody switched them off. In both cases, they keep drawing power, taking up rack space, and generating heat that has to be cooled. This waste is difficult for operators to detect, and what goes undetected is rarely corrected.

Traditional monitoring tools tell an administrator what a server is doing right now, but rarely surface whether that usage pattern justifies that resources being spent on it and almost never translates that pattern into an actionable recommendation.

A 2026 CEEW-SYSTEMIQ study puts India's data center capacity at roughly 1.5 GW, up nearly threefold since 2020 and headed toward 6.5 GW by 2030, already consuming close to 150 billion litres of water a year [1]. Yet most Indian states with data center policies don't even require PUE or WUE reporting, while the EU, Germany, China and Singapore all enforce some form of it [1]. That gap is the real problem here without a reporting requirement, the kind of waste that can sit unnoticed for years, largely because nothing in day-to-day operations forces anyone to go looking for it.

II. LITERATURE SURVEY

A. Energy-Aware Resource Allocation and Server Consolidation

Much of the early work on data center efficiency problems was allocation. Deciding where workloads should run so that fewer machines carry more of the load. Beloglazov and Buyya[2] proposed energy-aware heuristics for virtual machine placement and migration and showed that dynamic consolidation reduces power draw substantially compared to static allocation, though the system acts autonomously with no provision for human review before a decision is executed.

Subramanian et al. [3] approached the same problem with more mathematical precision, modeling server consolidation as a variant of variable-sized bin packing, proving it NP-hard and supplying an approximation algorithm with bounded guarantees, the tradeoff is an assumption of known, stable

demand that rarely holds in a data center where workloads shift hour to hour.

Two later studies complicate the picture usefully. [4] Lovasz et al. examined what consolidation costs in performance rather than only what it saves in energy. A tradeoff much of the purely energy-focused literature leaves unexamined.

[5] Uddin et al. then gave that tradeoff its concrete numbers, providing big drops in both power draw and total ownership cost after combining an enterprise server fleet, even though their study is a single retrospective audit rather than a system built to keep finding new inefficiencies as they happen.

B. Machine Learning-Based Workload and Resource Prediction

A more recent line of work uses machine learning to forecast resource use in advance, rather than optimizing allocation after the fact. [6] Bawa et al. built a two-stage ensemble with XGBoost and LightGBM. It first classifies whether a workload is idle or active, then predicts CPU and memory utilization for it.

[7] Deva and Jain took a similar approach, pairing Azure AutoML with XGBoost to predict cloud resource utilization in an enterprise setting, and they reported strong accuracy.

[8] Khan et al. went a step further by including provisioned resources, disk throughput and network metrics, which earlier prediction work mostly left out even though they clearly affect host-level energy use. One number from their study is worth citing directly, CPU and memory utilization in Google's own production clusters rarely went past 60%. That is exactly the kind of underutilization these systems are built to catch. Because measuring energy consumption at the level of individual virtual machines is difficult in practice, the same authors also proposed grouping machines by energy-relevant behaviour through clustering rather than instrumenting each one directly.

C. Carbon-Aware Scheduling and Cooling Optimization

[9] Radovanovic et al. describe a system used within Google that shifts flexible workloads across time and geography according to forecasted grid carbon intensity, cutting operational carbon meaningfully, though the approach depends on multi-region infrastructure and workload portability that exist mainly at hyperscale. Cooling receives its own dedicated treatment elsewhere.

[10] Zhan et al. trained a physics-informed offline reinforcement learning system on historical data specifically to avoid the risk of experimenting directly on a live cooling system and Alex et al. [11] combined reinforcement with agglomerative clustering to guide VM placement according to a carbon-efficiency score computed per data center. Neither reinforcement learning approach yet has been validated on production infrastructure. Both remain simulation results, promising but unproven at scale.

D. Sustainability Metrics, Policy and ESG Reporting

A different strand of the literature is less concerned with how to optimize a data center and more concerned with whether anyone is required to measure it all. [12] Grochulska-Salak et al.

argue that PUE alone cannot characterize sustainability and propose a broader set of indicators spanning water use, waste heat recovery, renewable integration and land use. Regulation is beginning to catch up with that argument; the EU's Corporate Sustainability Reporting Directive in force since January 2025, now requires large enterprises to disclose detailed environmental metrics and California's Climate Corporate Data Accountability Act (SB 253) will require Scope 1 and 2 emissions reporting starting in 2026 [13].

[14] Tripathi et al. offer a look at how this plays out on the ground. India's data center capacity has nearly tripled since 2020, yet most states have no PUE, WUE or carbon standards in their policies, which suggests the constraint is regulatory rather than technical. A 2025 perspective piece on what data center growth means for public health [15] found noise levels inside some facilities above safe exposure thresholds, though it notes that the evidence on these impacts is still thin overall. A similar pattern holds for water and energy disclosure: reporting from the Lincoln Institute of Land Policy [16] finds that operators routinely withhold consumption figures from host communities behind non-disclosure agreements. The engineering literature reaches a comparable conclusion by a different route. [17] A broader 2025 IEEE survey argues that energy efficiency, cooling and newer approaches such as hyper-converged infrastructure are most effective when pursued jointly rather than in isolation, a view anticipated by earlier work from Ahmed et al. [18], who showed that neural-network based workload forecasting improves energy efficiency without a corresponding loss in service quality.

E. Research Gap and Positioning of GreenOps

Across this body of work, three distinct strands emerge and each stops short of what the next one starts. Systems that optimize placement act on their own, with no step where a person reviews the decision before it executes. Systems that predict resource use stop at the forecast itself, without going on to rank or justify a corrective action. Work on sustainability reporting and policy documents the absence of continuous, standardized tracking but stops short of building anything to fill that gap. None of the systems reviewed here combine continuous ML-based prediction, rule-based waste detection and unified PUE/WUE/carbon reporting with a recommendation a human can actually review before anything changes. GreenOps is built at the point where these three strands meet.

III. METHODOLOGY

GreenOps is built as a continuous pipeline from raw telemetry to a human reviewed recommendation. No stage acts on infrastructure without explicit approval, unlike the optimization systems in Section II. Every stage writes back to a shared database for future reference. The system is organized into four functional modules, shown in Fig 1.

A. System Architecture

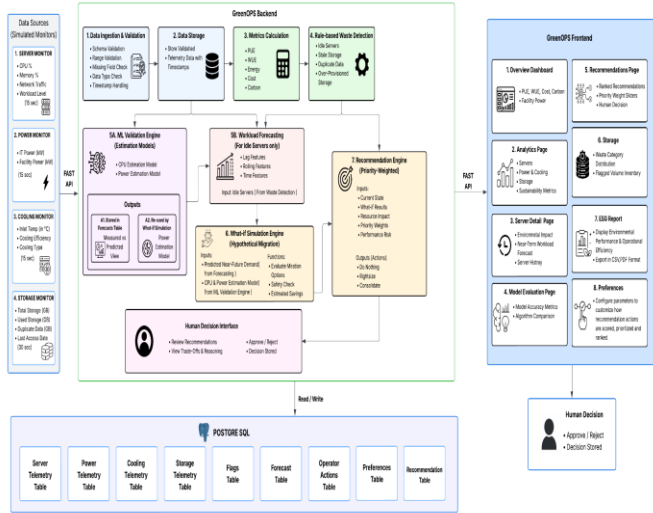


Fig. 1. GreenOps Architecture diagram

Fig 1. Shows the complete GreenOps architecture, spanning four layers: data source, backend processing, a shared persistence layer and a frontend dashboard. On the data-source side, four monitors, generate readings every 15 to 30 seconds - a Server Monitor tracking CPU, memory, network traffic and workload level; a Power Monitor tracking IT power and facility power; a Cooling Monitor tracking inlet temperature, cooling efficiency and cooling type and a Storage Monitor tracking total and used storage, duplicate data and days since last access. In this prototype, these four monitors are simulated rather than connected to physical hardware, since the working environment did not provide access to a live data center or its real monitoring tools. The simulators were built to mirror the structure and behavior of real telemetry sources so that the rest of the pipeline from ingestion through to recommendation, operates exactly as it would against a live facility. In a production deployment, this same ingestion layer would connect to the actual monitoring APIs that the data center already runs without requiring changes further down the pipeline.

Telemetry reaches the backend through a FastAPI ingestion layer, where it is checked for schema conformity, valid ranges and missing fields before being timestamped and written to a PostgreSQL database. From there, the same data feeds a metrics-calculation stage, a rule-based waste-detection stage and two machine learning stages; one estimating current CPU and power behavior and the other forecasting near-future demand for servers already flagged as idle. The forecasting output feeds a what-if simulation engine, which in turn feeds a priority-weighted recommendation engine. Every recommendation this engine produces is written to the database and surfaced through a React-based frontend, where an operator makes the final approve-or-reject decision through a human decision interface. Twelve distinct tables are in the shared database keeping every stage's inputs and outputs available to every other stage which is what allows recommendation

engines for instance, to reason about a server's full history rather than only its most recent reading.

B. Module 1: Data Pipeline and Observability Layer

This module covers ingestion, storage, real-time metrics calculation and rule-based waste detection and its job is to turn raw, noisy telemetry into something the rest of the system can trust. The live incoming data is being simulated currently since there is no access to real telemetry, the simulation runs multiple servers across four types: four Compute, three GPU, three Storage and three Edge servers with a few servers in each type deliberately biased toward low utilization, so the recommendation engine always has more than one real candidate to choose from.

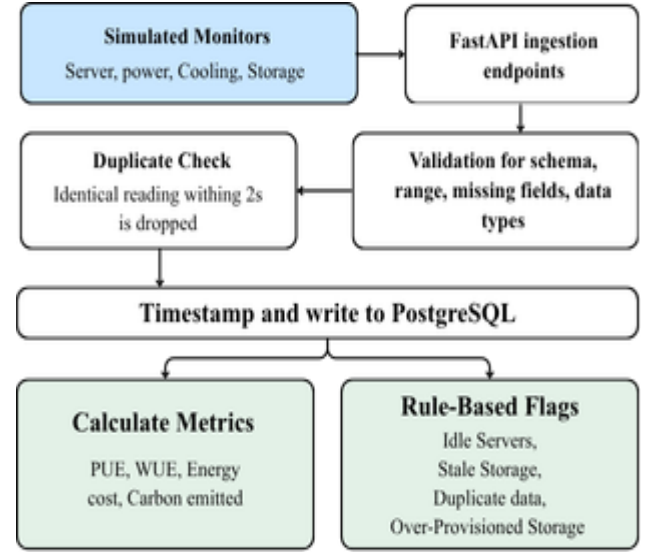


Fig. 2. Module 1 pipeline diagram

Fig 2. Shows how a reading gets from a simulated monitor to something the rest of the system can use. Server, power and cooling readings all post every 15 seconds. Storage posts every 30, since a volume's used space and duplicate count don't shift nearly as fast as a server's CPU does. Each reading goes through schema, range and type checks before it's timestamped and written to PostgreSQL and if an identical reading arrives within two seconds, it gets dropped so a retried delivery is never counted twice.

Once a reading clears validation it is sent to two places. It feeds the metrics calculation which produces PUE, WUE, energy cost and carbon emissions. Carbon uses a configurable grid emission factor defaulting to 0.710 kg/kWh based on India's national grid average and it can be adjusted for a different site.

$$PUE = \text{Facility Power (kW)} / \text{IT Power (kW)}$$

$$WUE = \text{Total Water consumed (L)} / \text{IT Energy Usage (kWh)}$$

$$\text{Energy Cost} = \text{Energy consumed} * \text{applied tariff}$$

$$\text{Carbon (kg)} = \text{Energy (kWh)} * 0.710 \text{ kg/kWh}$$

(a placeholder grid carbon intensity)

The reading is also fed into the rule engine, which checks the same reading for four waste patterns: idle servers when CPU usage is less than a threshold for a time period, stale

storage when data isn't being used in almost a year, duplicate data and over-provisioning where the resources used is lesser than the resource allocated.

C. Module 2: Predictive Intelligence Engine

This module covers CPU and power estimation, workload forecasting for idle servers and what-if simulation and its where GreenOps moves from describing the present state of the data center to reasoning about its near future. The three components are chained rather than independent: the estimation models establish what a server is doing right now, forecasting projects where an idle server is headed next and simulation uses both of these to test hypothetical changes before anything is recommended.

1) CPU and Power Estimation:

The ML validation engine trains two related models rather than just one.

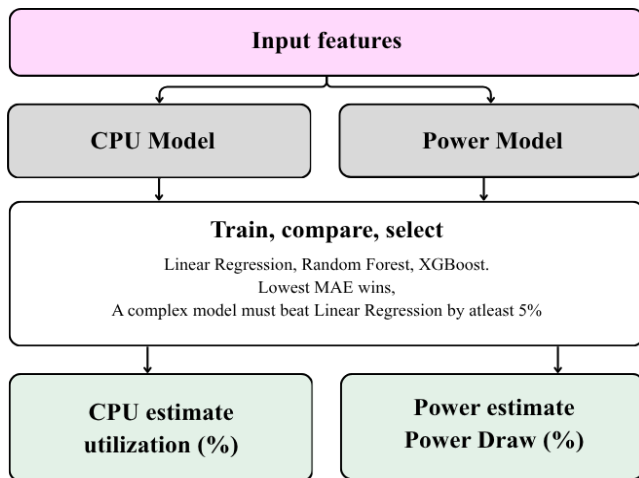


Fig. 3. CPU / Power estimation pipeline diagram

Fig 3. Shows the two estimation models - Power model and CPU model side by side. Both take the same categorical and numerical telemetry, server type, cooling type, region, time of day, workload, memory, network, inlet temperature, cooling efficiency and PUE. The power model gets one extra input, the server's own current CPU reading since it tracks how hard a server is working. Both models are trained and compared the same way using Linear regression, Random Forest and XGBoost algorithms. These three are trained with an 80/20 train-test split, scored on Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . The lowest Mean Absolute Error (MAE) wins and a complex model only gets chosen if it beats Linear Regression by more than 5%. On this dataset, Linear Regression won both times. A different environment could easily get Random Forest or XGBoost instead.

2) Workload Forecasting:

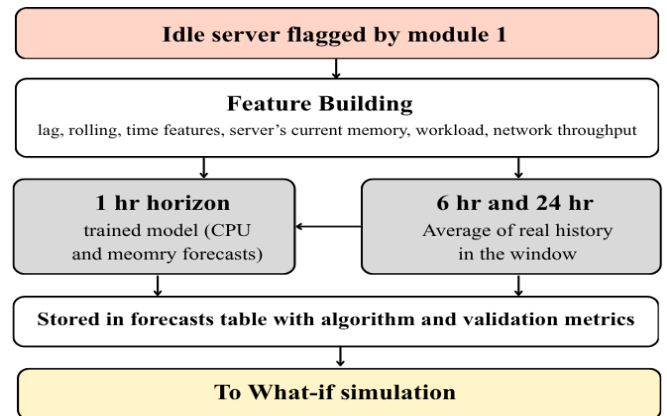


Fig. 4. Workload Forecasting pipeline diagram

Fig 4. Shows what happens to a server once Module 1 flags it as idle. GreenOps checks whether the server is genuinely underused or just between tasks right now. It does this by building three kinds of features from the server's own recent history: lag features for what it was doing a few readings ago, rolling features for short-term trends and time features for the daily rhythm most workloads follow. Its current memory, workload and network throughput are given as input too, so the model has both where the server has been and where it stands right now.

Two separate models come out of this, one for CPU demand and one for memory demand, each with its own file and its own MAE, RMSE and R^2 scores. This mirrors the CPU and power split, hence each target's accuracy and selected algorithm stay independently visible to an operator. Both forecasts feed the simulation engine when it evaluates a consolidation candidate's near-term demand for the server rather than reacting only to its most recent reading, which is what allows GreenOps to distinguish a server that is briefly quiet from the one that has been genuinely underutilized for an extended period.

At one hour out, there's enough real history to train a model properly. At six and twenty-four hours, a trained model would likely overshoot a server that's just holding steady. So GreenOps averages whatever real history actually exists in that window. If there isn't enough history yet, it says so instead of guessing. This helps keep the trust in this software since every number an operator sees traces back to something the system actually observed.

3) What-If Simulation:

This simulation gets the flagged server and the candidate server and asks: What would really happen if the server's workload was moved, combined or resized?

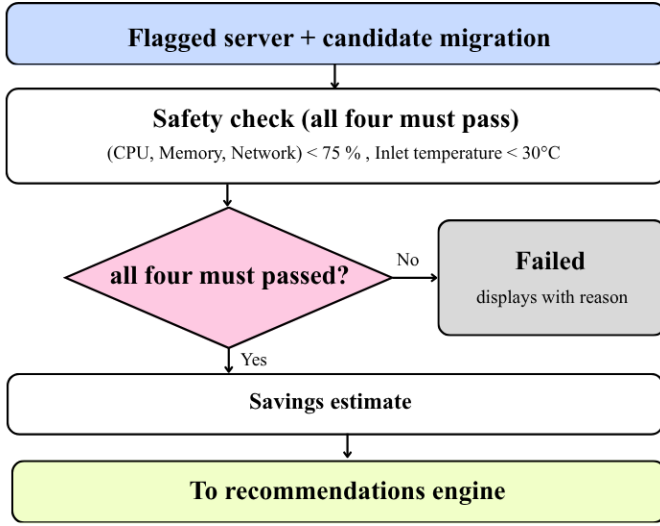


Fig. 5. What-if Simulation pipeline diagram

Fig 5. Shows the safety check every migration option has to pass. Before GreenOps estimates any savings, it asks whether moving a server’s workload is actually safe, not just whether it would save energy. A candidate has to clear four checks at once: CPU under 75%, memory under 75%, network throughput under 75% of link capacity and a predicted inlet temperature under 30°C after the move. CPU and memory get checked against both the current reading and the forecast, while network and thermal are judged on the current values only. A candidate only becomes the recommended target if it clears both the present-moment and the near-future check. But failing doesn’t mean disappearing. A candidate that fails just the forecast, the network limit or the thermal check still shows up to the operator with a specific reason attached.

Once a candidate clears all four checks, the engine estimates what the move would actually save:

$$\begin{aligned}
 \text{energy_saved} &= \max(0, \text{energy_before} - \text{energy_after}) \\
 \text{carbon_saved} &= \text{energy_saved} \times 0.710 \text{ kg/kWh} \\
 \text{cost_saved} &= \text{energy_saved} \times ₹8/\text{kWh} \\
 \text{water_saved} &= \text{energy_saved} \times \text{WUE_FACTOR} \\
 &\quad [\text{source's cooling type}]
 \end{aligned}$$

Energy_after comes from the power model’s prediction on the candidate’s full post-move profile, scaled to facility power via its own PUE, with a simple CPU-based estimate used as a fallback when the model or cooling data isn’t available.

D. Module 3: Decision Support and Human Interface

This module covers recommendation ranking, human review and the frontend dashboard and it is the layer where the prediction and simulations are turned into something an operator can actually act on. A priority-weighted recommendation engine combines four inputs - the server’s current state, the what-if simulation’s results, the estimated resource impact of each option and a set of configurable priority weights reflecting what the organization currently cares about most whether that is cost, carbon or performance risk, to rank pending recommendations against each other. Each

recommendation already carries a fixed action type determined by its originating flag such as rightsizing, consolidation, archival or deduplication and the operator then applies one of three decisions to any given recommendation: accept it, snooze it for re-evaluation or dismiss it. This ranking is computed as weighted sum of normalized savings, minus a penalty for risk:

$$\begin{aligned}
 \text{Score} &= w_{\text{energy}} \cdot \text{norm}(\text{energy_saved}) \\
 &+ w_{\text{cost}} \cdot \text{norm}(\text{cost_saved}) + w_{\text{carbon}} \cdot \text{norm}(\text{carbon_saved}) + \\
 &\quad w_{\text{water}} \cdot \text{norm}(\text{water_saved}) - w_{\text{risk}} \cdot \text{risk_score}
 \end{aligned}$$

Each value is normalized against the largest of its kind on the current batch, so raw units never dominate the score. Weights are stored per organization and editable live from the Preferences page, re-ranking recommendations instantly with no retraining. Rather than applying any of these automatically, the system writes each ranked recommendation to the database and surfaces it through the frontend and the IT infrastructure manager decides whether the recommendations are approved. Then the appropriate action is being performed meaning that no infrastructure change in GreenOps ever happened without a human decision behind it, the approved actions can also be viewed in a separate page to keep track of the actions. Every decision in GreenOps is also tied to who is allowed to make it. The system defines three roles: an infrastructure manager who can review, accept, snooze or reject recommendations and manage other users; a sustainability manager who has read-only access to the dashboards and ESG reports but cannot act on anything and an operations engineer who can mark an already-approved action as executed but cannot make the original accept-or-reject decision. This split helps differentiate roles of who is allowed to make the call, while execution and oversight are kept to separate roles. The thirteen-page React dashboard gives a complete overview of everything necessary for the users which is discussed in detail in Section IV.

E. Module 4: ESG Reporting

Alongside the operator-facing recommendation workflow, GreenOps maintains a dedicated ESG reporting layer that aggregates the telemetry into a facility-level sustainability rather than a per-decision view. This does not wait on human approval. Energy, carbon, cost, water, WUE, average PUE, average CPU and memory load, cooling efficiency and the count of underutilized servers are all calculated directly from raw telemetry over a selected date range, whether or not an operator has acted on anything during that window. A separate section of the same report tracks the decision side: how many recommendations were accepted, snoozed or dismissed and what the accepted ones were estimated to save in energy, carbon, cost and storage at the moment each was approved. Reports are downloadable in both CSV and PDF form.

IV. IMPLEMENTATION

The methodology described in Section III was implemented as a working prototype, covering the full pipeline from telemetry ingestion through to the human decision interface.

A. Backend Implementation

The backend is implemented in FastAPI with SQLAlchemy over PostgreSQL, exposing telemetry ingestion, analytics, forecasting, recommendation and ESG reporting endpoints

Each of the four simulated monitors post to its own ingestion endpoint, where they are validated before being written to their respective telemetry table.

TABLE I. DATABASE BACKEND

No	Table	Purpose
1.	Servers	Server registry - static identity / metadata for each of the 13 simulated servers
2.	Server telemetry	Raw compute readings, posted every 15s
3.	Power telemetry	Raw power readings, posted every 15s
4.	Cooling telemetry	Raw cooling readings, posted every 15s
5.	Storage telemetry	Raw storage readings posted every 30s
6.	Forecasts	Persisted CPU forecasts (trained model at one hour, historical average at six and twenty-four hours) with each prediction's algorithm and validation metrics.
7.	Flags	Active / cleared rule based waste detection (idle, stale, duplicate, over provisioned)
8.	Recommendations	Ranked actions generated from a flag, decision-support only
9.	Operator actions	The operator's decision and the what-if impact estimate at that exact moment
10.	Preferences	Single-row, live editable scoring weights used by the recommendation ranking formula
11.	Users	To store Role based access and sign in accordingly.
12.	Site Settings	Single-row, operator-editable config - grid emission factor plus declared values telemetry can't derive (installed capacity, renewable/reuse factors)

B. Frontend Implementation

The frontend is a React single-page application with thirteen routed pages.

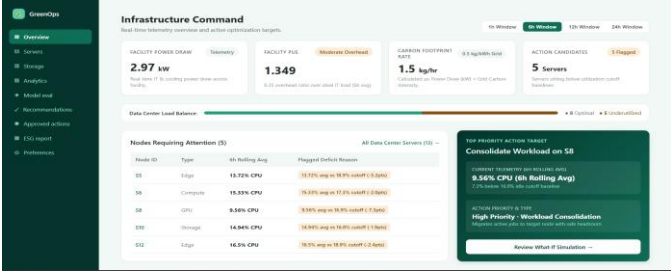


Fig. 6. Dashboard Overview page

The Overview page summarizes facility wide values for PUE, cumulative energy, carbon and the current count of flagged servers being refreshed from live simulated telemetry.

The Servers page shows the full server inventory filterable by type, region and status (Healthy/Underutilized/Stale/No Data), giving an operator a single screen to scan everything before drilling into any one machine. The Server detail page gives insights into per-server view combining live

environmental reading, the CPU forecast and any active waste flags for that specific server.

The Storage page shows specific waste view, surfacing stale, duplicated and over-provisioned volumes flagged by the rule engine.

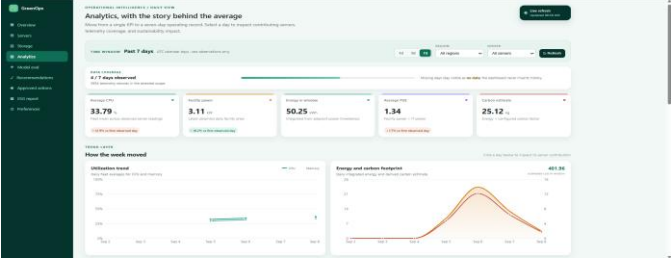


Fig. 7. Analytics page

The Analytics page breaks down the facility-level of idle servers, power/cooling performance and storage waste over time which is used to spot trends.

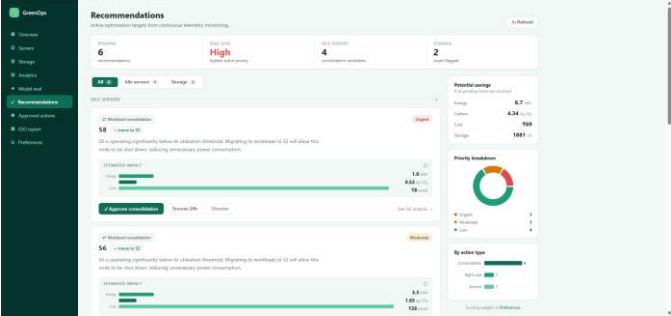


Fig. 8. Recommendations page

The Recommendations page ranks a list of pending consolidate/rightsized actions with inline Accept / Snooze / Dismiss controls, mini impact charts and a sidebar summarizing pending savings across the batch. The Recommendation details page gives full insights about a single recommendation. It ranks all the candidate tables with a CPU history chart, a storage breakdown and the ability to switch candidates and see the impact estimate re-scale live.

The Approved actions logs the past operator's decisions, filterable by executed vs awaiting executions where an operator can retroactively attach an executed at timestamp and a note once the action is carried out.

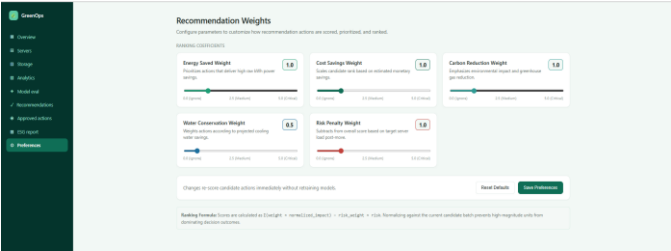


Fig. 9. Preferences page

The Preferences page has a weight slider for cost, carbon, water and risk that feeds directly into the recommendation scoring formula, re-ranking pending recommendations instantly without retraining any model.

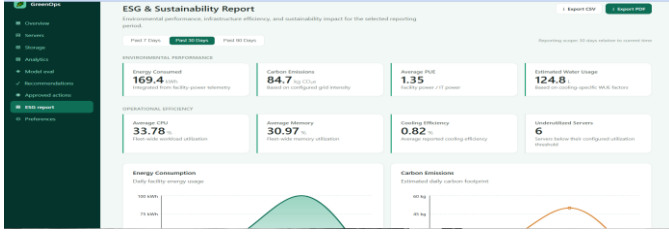


Fig. 10. ESG reporting page 1

This page has aggregated log of operator-action over a selected date range, exportable as CSV or PDF, presented explicitly as a decision record.

V. RESULTS AND EVALUATION

A. CPU and Power Estimation

Both estimation models were trained on the Kaggle “Green AI Data Center Telemetry” dataset (10,000 records and 15 columns), split by 80/20 with a most-frequent category imputer handling missing values in data center region (3,990 missing entries). The selection rule kept Linear Regression as the default and only promoted Random Forest or XGBoost if either beat it by more than 5% relative MAE.

TABLE II. CPU ESTIMATION

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	4.511	7.671	0.845
Random Forest	4.828	8.077	0.828
XGBoost	4.833	7.999	0.831

TABLE III. POWER ESTIMATION

Model	MAE (kW)	RMSE (kW)	R^2
Linear Regression	0.211	0.330	0.422
Random Forest	0.227	0.346	0.366
XGBoost	0.222	0.341	0.385

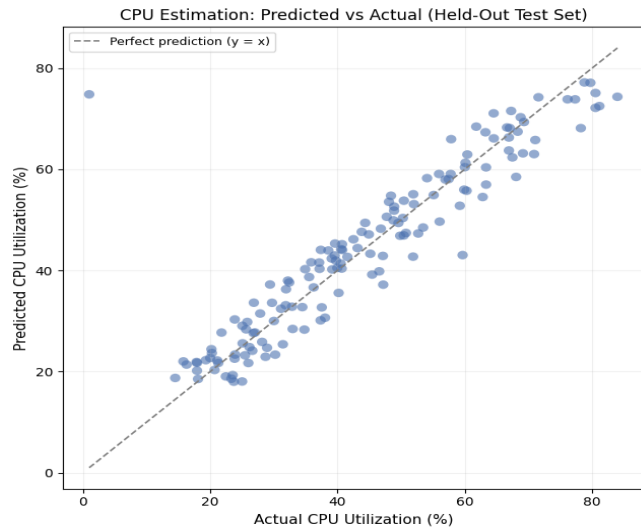


Fig. 11. CPU Estimation: Predicted vs Actual

Fig.11 demonstrates the practical accuracy of our model, where an R^2 value of 0.845 shows a strong alignment between predicted and actual CPU utilization. The data points consistently follow the ideal prediction line, with minor variance isolated to the extreme values where training samples were naturally sparse.

B. Workload Forecasting

The forecasting models were trained separately using telemetry accumulated by the live simulator.

TABLE IV. CPU WORKLOAD FORECAST

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	6.514	8.090	0.886
Random Forest	6.703	8.392	0.877
XGBoost	6.567	8.178	0.883

TABLE V. MEMORY WORKLOAD FORECAST

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	6.560	8.185	0.851
Random Forest	6.843	8.527	0.839
XGBoost	6.795	8.489	0.840

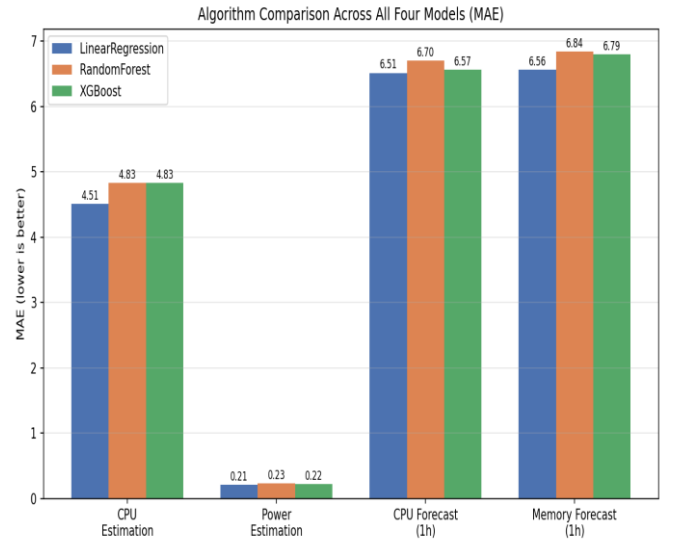


Fig. 12. Comparing algorithms across all four models using MAE

Across all three algorithms the margin Linear Regression won all four targets by the selection rule’s 5% margin. So the simpler model was kept throughout as shown in Fig. 12.

VI. CONCLUSION AND FUTURE WORK

GreenOps demonstrates that the capabilities most of the reviewed literature treats separately with continuous observability, ML-based prediction, rule-based waste detection and sustainability reporting can run as one pipeline with a single human decision point at the end. The working prototype validates this end to end with estimation models trained and honestly evaluated on real telemetry, forecasting models trained

on the system's own accumulated history, a four-dimension what-if safety check that screens every migration option before it's ever surfaced and role-based review that ties each decision to a specific accountable person rather than to automation.

Future work will focus on replacing the simulated telemetry with real integrations (Prometheus, smart PDU's/BMS, API's) and swapping the placeholder carbon-intensity and WUE constants for live regional grid and facility water data. The what-if check should also gain workload criticality gating, along with SLA requirements and storage and electrical capacity limits, so that no recommendation is surfaced for a workload that is too critical to move. Longer-range planning needs a 24-hour forecast. That's not possible yet, since there isn't enough telemetry to train it on. Nothing currently checks whether an approved migration saved what was predicted either, so a before-and-after comparison of power readings is worth adding. Beyond that, GreenOps could be tried across several data centers and watched over a longer period, to see whether operators come to trust it as its track record grows.

VII. REFERENCES

- [1] Council on Energy, Environment and Water (CEEW), "How is data centre infrastructure in India shaping power and water use?," 2026.
- [2] A. Beloglazov, J. Abawajy and R. Buyya, "Energy-aware resource allocation heuristics for efficient management of data centers for cloud computing," *Future Generation Computer Systems*, vol. 28, no. 5, pp. 755–768, 2012.
- [3] C. Subramanian, A. Vasan and A. Sivasubramaniam, "Reducing data center power with server consolidation: Approximation and evaluation," in *Proc. 17th Int. Conf. High Performance Computing (HiPC)*, IEEE Computer Society, 2010.
- [4] G. Lovász, F. Niedermeier and H. de Meer, "Performance tradeoffs of energy-aware virtual machine consolidation," *Cluster Computing*, vol. 16, no. 3, pp. 481–496, 2013.
- [5] M. Uddin *et al.*, "Server consolidation: A technique to enhance cloud data center power efficiency and overall cost of ownership," *International Journal of Distributed Sensor Networks*, 2021.
- [6] J. Bawa, K. K. Chahal and K. Kaur, "Improving cloud resource management: An ensemble learning approach for workload prediction," *The Journal of Supercomputing*, vol. 81, no. 10, 2025.
- [7] M. R. R. Deva and N. Jain, "Utilizing Azure Automated Machine Learning and XGBoost for predicting cloud resource utilization in enterprise environments," in *Proc. 2025 Int. Conf. Networks and Cryptology (NETCRYPT)*, IEEE, 2025.
- [8] T. Khan, W. Tian, S. Ilager and R. Buyya, "Workload forecasting and energy state estimation in cloud data centres: ML-centric approach," *Future Generation Computer Systems*, vol. 128, pp. 320–332, 2022.
- [9] A. Radovanović *et al.*, "Carbon-aware computing for datacenters," arXiv preprint arXiv:2106.11750, 2021.
- [10] X. Zhan *et al.*, "Data center cooling system optimization using offline reinforcement learning," arXiv preprint arXiv:2501.15085, 2025.
- [11] M. Alex, S. O. Ojo and F. M. Awuor, "Carbon-aware, energy-efficient and SLA-compliant virtual machine placement in cloud data centers using deep Q-networks and agglomerative clustering," *Computers*, vol. 14, no. 7, Art. no. 280, 2025.
- [12] M. Grochulska-Salak *et al.*, "Green data centres: Sustainable solutions with green energy and green-blue infrastructure," *Energies*, vol. 18, no. 24, Art. no. 6592, 2025.
- [13] European Commission, "Corporate sustainability reporting," Financial Markets, Company Reporting and Auditing.
- [14] Council on Energy, Environment and Water (CEEW) and SYSTEMIQ, "Scaling India's data centre ecosystem," 2026.
- [15] Y. Tao and P. Gao, "Global data center expansion and human health: A call for empirical research," *Eco-Environment & Health*, vol. 4, no. 3, Art. no. 100157, 2025.
- [16] J. Gorey, "Data drain: The land and water impacts of the AI boom," *Land Lines Magazine*, Lincoln Institute of Land Policy, Oct. 17, 2025.
- [17] V. Vaccaro, L. C. Tagliabue and M. Aldinucci, "Sustainable data centers: Advancing energy efficiency and resource optimization," in *Proc. 33rd Euromicro Int. Conf. Parallel, Distributed and Network-Based Processing (PDP)*, Turin, Italy, 2025, pp. 486–493.
- [18] A. Ahmed, D. Brown and A. Gegov, "Dynamic resource allocation through workload prediction for energy efficient computing," in *Advances in Computational Intelligence Systems*, P. Angelov, A. Gegov, C. Jayne and Q. Shen, Eds., Springer, 2017, pp. 35–44.

Submission Summary

Conference Name

13th International Conference on Advanced Computing - ICOAC2026

Paper ID

285

Paper Title

Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization

Abstract

On-premises data centers today operate around the clock to guarantee availability but a large share of the electricity, cooling capacity and water they have to consume goes toward servers that are barely being used. This stays hidden because the telemetry that could reveal it are typically scattered across disconnected monitoring tools that were never built to talk to each other. Our work addresses this gap through GreenOps, an observability and resource intelligence platform that unifies this data in one integrated pipeline and makes it intelligible to human operators. The system collects live telemetry (simulated in this prototype) from server, power, cooling and storage monitors and stores it with full historical context and computes real time sustainability metrics including Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), energy cost and carbon emissions. GreenOps employs four machine learning models - CPU estimation, power estimation, workload forecasting and memory workload forecasting each trained from related telemetry. For every model, three candidate algorithms (Linear Regression, Random Forest and XGBoost) are evaluated honestly on held out data giving operators both a real-time and a near-future predictive view. Idle servers, stale storage and over-provisioned resources get flagged by a rule-based detection layer. If a server is flagged idle, a forecasting layer projects its near term and a what-if simulation engine trials possible migrations and estimates the savings involved. A priority-weighted recommendation engine then ranks these against cost, carbon and performance risk. A React based dashboard surfaces all of this, built for a human to review rather than for automation to act on unsupervised. This paper describes the system's architecture, the reasoning behind the modeling choices and results from our working prototype.

Created

2026-10-02 14:05:24 GMT+5:30

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Submission Files

Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization.pdf (910.5 Kb, 2026-10-02 14:01:28 GMT+5:30)

Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization

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Abstract— On-premises data centers today operate around the clock to guarantee availability but a large share of the electricity, cooling capacity and water they have to consume goes toward servers that are barely being used. This stays hidden because the telemetry that could reveal it are typically scattered across disconnected monitoring tools that were never built to talk to each other. Our work addresses this gap through GreenOps, an observability and resource intelligence platform that unifies this data in one integrated pipeline and makes it intelligible to human operators. The system collects live telemetry (simulated in this prototype) from server, power, cooling and storage monitors and stores it with **5** full historical context and computes real time sustainability metrics including Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), energy cost and carbon emissions. GreenOps employs four machine learning models - CPU estimation, power estimation, workload forecasting and memory workload forecasting each trained from related telemetry. For every model, three candidate algorithms (Linear Regression, Random Forest and XGBoost) are evaluated honestly on held out data giving operators both a real-time and a near-future predictive view. Idle servers, stale storage and over-provisioned resources get flagged by a rule-based detection layer. If a server is flagged idle, a forecasting layer projects its near term and a what-if simulation engine trials possible migrations and estimates the savings involved. A priority-weighted recommendation engine then ranks these against cost, carbon and performance risk. A React based dashboard surfaces all of this, built for a human to review rather than for automation to act on unsupervised. This paper describes the system's architecture, the reasoning behind the modeling choices and results from our working prototype.

Keywords— GreenOps, On-premises Data Centers, Sustainability, Machine Learning, Resource Optimization, Workload Forecasting, PUE, WUE

I. INTRODUCTION

Data Centers have quietly become one of the most power-hungry categories of infrastructure in the modern world. Whether a website is being loaded, a model is being trained, or a file is being synced to the cloud, it all traces back to a room full of servers running continuously throughout the day, regardless of whether anyone is actually using them at that moment. This 24/7 operation is by design, as availability is the goal, but it occurs at a steep environmental cost. The facilities that support this architecture also carry enormous overhead in the form of cooling systems, backup power infrastructure and

water usage all of which is driven by how much hardware is kept running rather than by how useful the work it performs is.

A significant portion of this consumption is completely avoidable. Servers frequently sit at low utilization for long periods of time, often because they were sized for a peak that rarely arrives, or because the workload they were built for moved on and nobody switched them off. In both cases, they keep drawing power, taking up rack space, and generating heat that has to be cooled. This waste is difficult for operators to detect, and what goes undetected is rarely corrected.

Traditional monitoring tools tell an administrator what a server is doing right now, but rarely surface whether that usage pattern justifies that resources being spent on it and almost never translates that pattern into an actionable **11** recommendation.

A 2026 CEEW-SYSTEMIQ study puts India's data center capacity at roughly **1.5 GW**, up nearly threefold since 2020 and headed toward 6.5 GW by 2030, already consuming close to 150 billion litres of water a year [1]. Yet most Indian states with data center policies don't even require PUE or WUE reporting, while the EU, Germany, China and Singapore all enforce some form of it [1]. That gap is the real problem here without a reporting requirement, the kind of waste that can sit unnoticed for years, largely because nothing in day-to-day operations forces anyone to go looking for it.

II. LITERATURE SURVEY

A. Energy-Aware Resource Allocation and Server Consolidation

Much of the early work on data center efficiency problems was allocation. Deciding where workloads should run so that fewer machines carry more of the load **7** Beloglazov and Buyya[2] proposed energy-aware heuristics for virtual machine placement and migration and showed that dynamic consolidation reduces power draw substantially compared to static allocation, though the system acts autonomously with no provision for human review before a decision is executed.

Subramanian et al. [3] approached the same problem with **10** more mathematical precision, modeling server consolidation as a variant of variable-sized bin packing, proving it NP-hard and supplying an approximation algorithm with bounded guarantees, the tradeoff is an assumption of known, stable

demand that rarely holds in a data center where workloads shift hour to hour.

Two later studies complicate the picture usefully. [4] Lovasz et al. examined what consolidation costs in performance rather than only what it saves in energy. A tradeoff much of the purely energy-focused literature leaves unexamined.

[5] Uddin et al. then gave that tradeoff its concrete numbers, providing big drops in both power draw and total ownership cost after combining an enterprise server fleet, even though their study is a single retrospective audit rather than a system built to keep finding new inefficiencies as they happen.

B. Machine Learning-Based Workload and Resource Prediction

A more recent line of work uses machine learning to forecast resource use in advance, rather than optimizing allocation after the fact. [6] Bawa et al. built a two-stage ensemble with XGBoost and LightGBM. It first classifies whether a workload is idle or active, then predicts CPU and memory utilization for it.

[7] Deva and Jain took a similar approach, pairing **Azure AutoML** with **XGBoost** to predict cloud resource utilization in an enterprise setting, and they reported strong accuracy.

[8] Khan et al. went a step further by including provisioned resources, disk throughput and network metrics, which earlier prediction work mostly left out even though they clearly affect host-level energy use. One number from their study is worth citing directly, CPU and memory utilization in Google's own production clusters rarely went past 60%. That is exactly the kind of underutilization these systems are built to catch. Because measuring energy consumption at the level of individual virtual machines is difficult in practice, the same authors also proposed grouping machines by energy-relevant behaviour through clustering rather than instrumenting each one directly.

C. Carbon-Aware Scheduling and Cooling Optimization

[9] Radovanovic et al. describe a system used within Google that shifts flexible workloads across time and geography according to forecasted grid carbon intensity, cutting operational carbon meaningfully, though the approach depends on multi-region infrastructure and workload portability that exist mainly at hyperscale. Cooling receives its own dedicated treatment elsewhere.

[10] Zhan et al. trained a physics-informed offline reinforcement learning system on historical data specifically to avoid the risk of experimenting directly on a live cooling system and Alex et al. [11] combined reinforcement with agglomerative clustering to guide VM placement according to a carbon-efficiency score computed per data center. Neither reinforcement learning approach yet has been validated on production infrastructure. Both remain simulation results, promising but unproven at scale.

D. Sustainability Metrics, Policy and ESG Reporting

A different strand of the literature is less concerned with how to optimize a data center and more concerned with whether anyone is required to measure it all. [12] Grochulska-Salak et al.

argue that PUE alone cannot characterize sustainability and propose a broader set of indicators spanning water use, waste heat recovery, renewable integration and **land use**. Regulation is beginning to catch up with that argument; **the EU's Corporate Sustainability Reporting Directive** in force since January 2025, now requires large enterprises to disclose detailed environmental metrics **and California's Climate Corporate Data Accountability Act (SB 253)** will require **Scope 1 and 2 emissions reporting starting in 2026** [13].

[14] Tripathi et al. offer a look at how this plays out on the ground. India's data center capacity has nearly tripled since 2020, yet most states have no PUE, WUE or carbon standards in their policies, which suggests the constraint is regulatory rather than technical. A 2025 perspective piece on what data center growth means for public health [15] found noise levels inside some facilities above safe exposure thresholds, though it notes that the evidence on these impacts is still thin overall. A similar pattern holds for water and energy disclosure: reporting from the Lincoln Institute of Land Policy [16] finds that operators routinely withhold consumption figures from host communities behind non-disclosure agreements. The engineering literature reaches a comparable conclusion by a different route. [17] A broader 2025 IEEE survey argues that energy efficiency, cooling and newer approaches such as hyper-converged infrastructure are most effective when pursued jointly rather than in isolation, a view anticipated by earlier work from Ahmed et al. [18], who showed that neural-network based workload forecasting improves energy efficiency without a corresponding loss in service quality.

E. Research Gap and Positioning of GreenOps

Across this body of work, three distinct strands emerge and each stops short of what the next one starts. Systems that optimize placement act on their own, with no step where a person reviews the decision before it executes. Systems that predict resource use stop at the forecast itself, without going on to rank or justify a corrective action. Work on sustainability reporting and policy documents the absence of continuous, standardized tracking but stops short of building anything to fill that gap. None of the systems reviewed here combine continuous ML-based prediction, rule-based waste detection and unified PUE/WUE/carbon reporting with a recommendation a human can actually review before anything changes. GreenOps is built at the point where these three strands meet.

III. METHODOLOGY

GreenOps is built as a continuous pipeline from raw telemetry to a human reviewed recommendation. No stage acts on infrastructure without explicit approval, unlike the optimization systems in Section II. Every stage writes back to a shared database for future reference. **The system is organized into four functional modules**, shown in Fig 1.

3 A. System Architecture

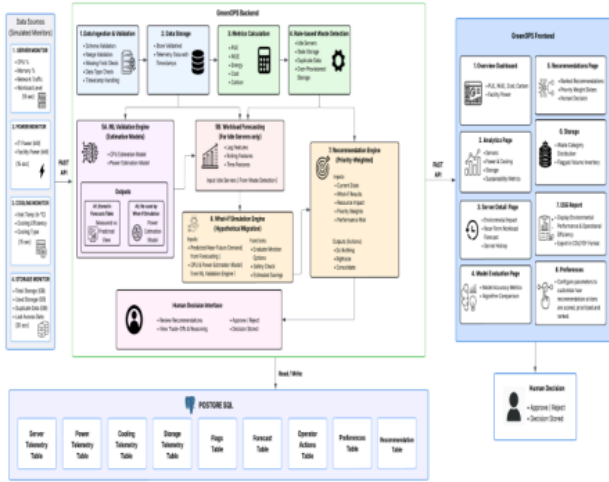


Fig. 1. GreenOps Architecture diagram

Fig 1. Shows the complete GreenOps architecture, spanning four layers: data source, backend processing, a shared persistence layer and a frontend dashboard. On the data-source side, four monitors, generate readings every 15 to 30 seconds - a Server Monitor tracking CPU, memory, network traffic and workload level; a Power Monitor tracking IT power and facility power; a Cooling Monitor tracking inlet temperature, cooling efficiency and cooling type and a Storage Monitor tracking total and used storage, duplicate data and days since last access. In this prototype, these four monitors are simulated rather than connected to physical hardware, since the working environment did not provide access to a live data center or its real monitoring tools. The simulators were built to mirror the structure and behavior of real telemetry sources so that the rest of the pipeline from ingestion through to recommendation, operates exactly as it would against a live facility. In a production deployment, this same ingestion layer would connect to the actual monitoring APIs that the data center already runs without requiring changes further down the pipeline.

Telemetry reaches the backend through a FastAPI ingestion layer, where it is checked for schema conformity, valid ranges and missing fields before being timestamped and written to a PostgreSQL database. From there, the same data feeds a metrics-calculation stage, a rule-based waste-detection stage and two machine learning stages; one estimating current CPU and power behavior and the other forecasting near-future demand for servers already flagged as idle. The forecasting output feeds a what-if simulation engine, which in turn feeds a priority-weighted recommendation engine. Every recommendation this engine produces is written to the database and surfaced through a React-based frontend, where an operator makes the final approve-or-reject decision through a human decision interface. Twelve distinct tables are in the shared database keeping every stage's inputs and outputs available to every other stage which is what allows recommendation

engines for instance, to reason about a server's full history rather than only its most recent reading.

B. Module 1: Data Pipeline and Observability Layer

This module covers ingestion, storage, real-time metrics calculation and rule-based waste detection and its job is to turn raw, noisy telemetry into something the rest of the system can trust. The live incoming data is being simulated currently since there is no access to real telemetry, the simulation runs multiple servers across four types: four Compute, three GPU, three Storage and three Edge servers with a few servers in each type deliberately biased toward low utilization, so the recommendation engine always has more than one real candidate to choose from.

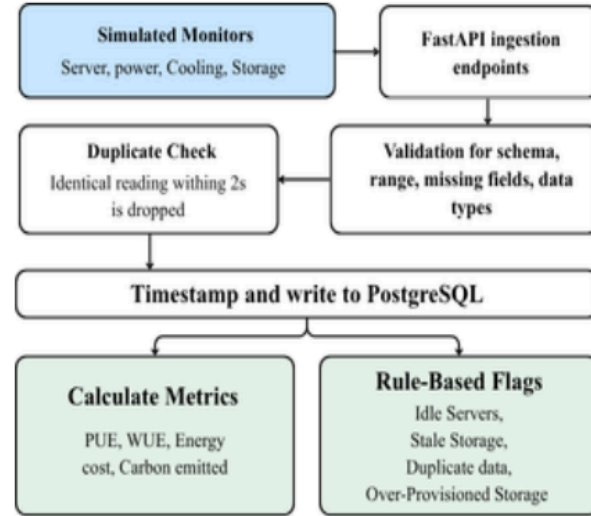


Fig. 2. Module 1 pipeline diagram

Fig 2. Shows how a reading gets from a simulated monitor to something the rest of the system can use. Server, power and cooling readings all post every 15 seconds. Storage posts every 30, since a volume's used space and duplicate count don't shift nearly as fast as a server's CPU does. Each reading goes through schema, range and type checks before it's timestamped and written to PostgreSQL and if an identical reading arrives within two seconds, it gets dropped so a retried delivery is never counted twice.

Once a reading clears validation it is sent to two places. It feeds the metrics calculation which produces PUE, WUE, energy cost and carbon emissions. Carbon uses a configurable grid emission factor defaulting to 0.710 kg/kWh based on India's national grid average and it can be adjusted for a different site.

$$PUE = \text{Facility Power (kW)} / \text{IT Power (kW)}$$

$$WUE = \text{Total Water consumed (L)} / \text{IT Energy Usage (kWh)}$$

$$\text{Energy Cost} = \text{Energy consumed} * \text{applied tariff}$$

$$\text{Carbon (kg)} = \text{Energy (kWh)} * 0.710 \text{ kg/kWh}$$

(a placeholder grid carbon intensity)

The reading is also fed into the rule engine, which checks the same reading for four waste patterns: idle servers when CPU usage is less than a threshold for a time period, stale

storage when data isn't being used in almost a year, duplicate data and over-provisioning where the resources used is lesser than the resource allocated.

C. Module 2: Predictive Intelligence Engine

This module covers CPU and power estimation, workload forecasting for idle servers and what-if simulation and its where GreenOps moves from describing the present state of the data center to reasoning about its near future. The three components are chained rather than independent: the estimation models establish what a server is doing right now, forecasting projects where an idle server is headed next and simulation uses both of these to test hypothetical changes before anything is recommended.

1) CPU and Power Estimation:

The ML validation engine trains two related models rather than just one.

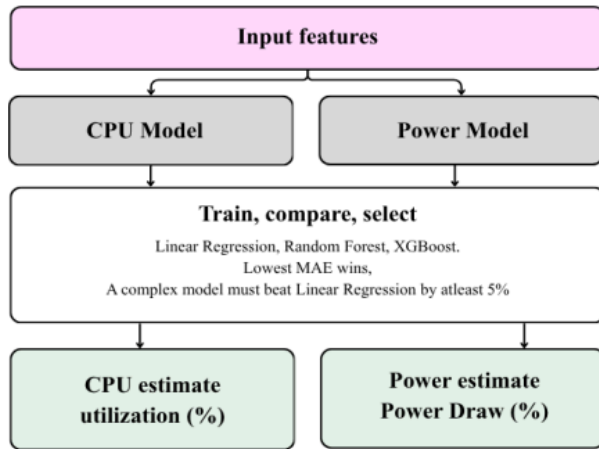


Fig. 3. CPU / Power estimation pipeline diagram

Fig 3. Shows the two estimation models - Power model and CPU model side by side. Both take the same categorical and numerical telemetry, server type, cooling type, region, time of day, workload, memory, network, inlet temperature, cooling efficiency and PUE. The power model gets one extra input, the server's own current CPU reading since it tracks how hard a server is working. Both models are trained and compared the same way using Linear regression, Random Forest and XGBoost algorithms. The three are trained with an 80/20 train-test split, scored on Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . The lowest Mean Absolute Error (MAE) wins and a complex model only gets chosen if it beats Linear Regression by more than 5%. On this dataset, Linear Regression won both times. A different environment could easily get Random Forest or XGBoost instead.

2) Workload Forecasting:

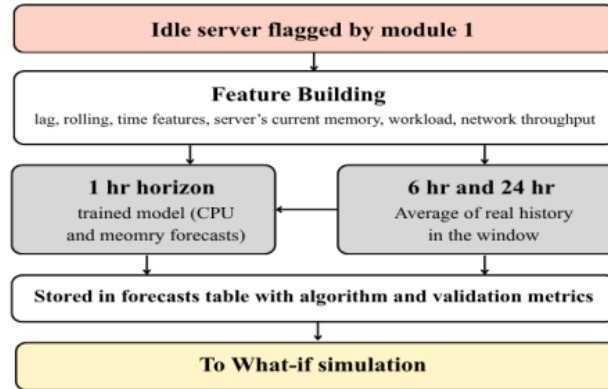


Fig. 4. Workload Forecasting pipeline diagram

Fig 4. Shows what happens to a server once Module 1 flags it as idle. GreenOps checks whether the server is genuinely underused or just between tasks right now. It does this by building three kinds of features from the server's own recent history: lag features for what it was doing a few readings ago, rolling features for short-term trends and time features for the daily rhythm most workloads follow. Its current memory, workload and network throughput are given as input too, so the model has both where the server has been and where it stands right now.

Two separate models come out of this, one for CPU demand and one for memory demand, each with its own file and its own MAE, RMSE and R^2 scores. This mirrors the CPU and power split, hence each target's accuracy and selected algorithm stay independently visible to an operator. Both forecasts feed the simulation engine when it evaluates a consolidation candidate's near-term demand for the server rather than reacting only to its most recent reading, which is what allows GreenOps to distinguish a server that is briefly quiet from the one that has been genuinely underutilized for an extended period.

At one hour out, there's enough real history to train a model properly. At six and twenty-four hours, a trained model would likely overshoot a server that's just holding steady. So GreenOps averages whatever real history actually exists in that window. If there isn't enough history yet, it says so instead of guessing. This helps keep the trust in this software since every number an operator sees traces back to something the system actually observed.

3) What-If Simulation:

This simulation gets the flagged server and the candidate server and asks: What would really happen if the server's workload was moved, combined or resized?

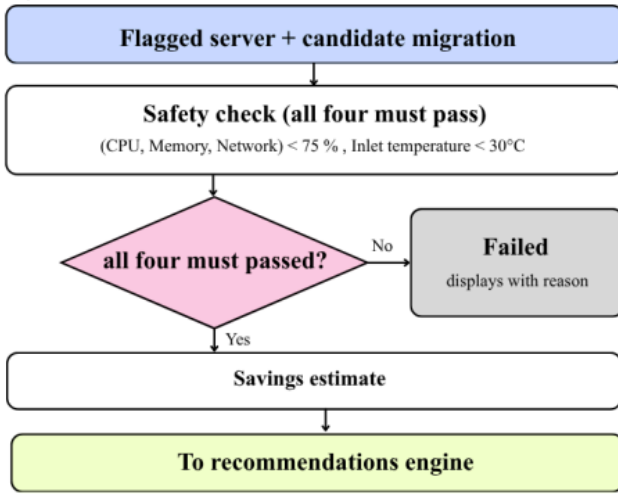


Fig. 5. What-if Simulation pipeline diagram

Fig 5. Shows the safety check every migration option has to pass. Before GreenOps estimates any savings, it asks whether moving a server's workload is actually safe, not just whether it would save energy. A candidate has to clear four checks at once: CPU under 75%, memory under 75%, network throughput under 75% of link capacity and a predicted inlet temperature under 30°C after the move. CPU and memory get checked against both the current reading and the forecast, while network and thermal are judged on the current values only. A candidate only becomes the recommended target if it clears both the present-moment and the near-future check. But failing doesn't mean disappearing. A candidate that fails just the forecast, the network limit or the thermal check still shows up to the operator with a specific reason attached.

Once a candidate clears all four checks, the engine estimates what the move would actually save:

$$\begin{aligned}
 \text{energy_saved} &= \max(0, \text{energy_before} - \text{energy_after}) \\
 \text{carbon_saved} &= \text{energy_saved} \times 0.710 \text{ kg/kWh} \\
 \text{cost_saved} &= \text{energy_saved} \times ₹8/\text{kWh} \\
 \text{water_saved} &= \text{energy_saved} \times \text{WUE_FACTOR} \\
 &\quad [\text{source's cooling type}]
 \end{aligned}$$

Energy_after comes from the power model's prediction on the candidate's full post-move profile, scaled to facility power via its own PUE, with a simple CPU-based estimate used as a fallback when the model or cooling data isn't available.

D. Module 3: Decision Support and Human Interface

This module covers recommendation ranking, human review and the frontend dashboard and it is the layer where the prediction and simulations are turned into something an operator can actually act on. A priority-weighted recommendation engine combines four inputs - the server's current state, the what-if simulation's results, the estimated resource impact of each option and a set of configurable priority weights reflecting what the organization currently cares about most whether that is cost, carbon or performance risk, to rank pending recommendations against each other. Each

recommendation already carries a fixed action type determined by its originating flag such as rightsizing, consolidation, archival or deduplication and the operator then applies one of three decisions to any given recommendation: accept it, snooze it for re-evaluation or dismiss it. This ranking is computed as weighted sum of normalized savings, minus a penalty for risk:

$$\begin{aligned}
 \text{Score} &= w_{\text{energy}} \cdot \text{norm}(\text{energy_saved}) \\
 &+ w_{\text{cost}} \cdot \text{norm}(\text{cost_saved}) + w_{\text{carbon}} \cdot \text{norm}(\text{carbon_saved}) + \\
 &\quad w_{\text{water}} \cdot \text{norm}(\text{water_saved}) - w_{\text{risk}} \cdot \text{risk_score}
 \end{aligned}$$

Each value is normalized against the largest of its kind on the current batch, so raw units never dominate the score. Weights are stored per organization and editable live from the Preferences page, re-ranking recommendations instantly with no retraining. Rather than applying any of these automatically, the system writes each ranked recommendation to the database and surfaces it through the frontend and the IT infrastructure manager decides whether the recommendations are approved. Then the appropriate action is being performed meaning that no infrastructure change in GreenOps ever happened without a human decision behind it, the approved actions can also be viewed in a separate page to keep track of the actions. Every decision in GreenOps is also tied to who is allowed to make it. The system defines three roles: an infrastructure manager who can review, accept, snooze or reject recommendations and manage other users; a sustainability manager who has read-only access to the dashboards and ESG reports but cannot act on anything and an operations engineer who can mark an already-approved action as executed but cannot make the original accept-or-reject decision. This split helps differentiate roles of who is allowed to make the call, while execution and oversight are kept to separate roles. The thirteen-page React dashboard gives a complete overview of everything necessary for the users which is discussed in detail in Section IV.

E. Module 4: ESG Reporting

Alongside the operator-facing recommendation workflow, GreenOps maintains a dedicated ESG reporting layer that aggregates the telemetry into a facility-level sustainability rather than a per-decision view. This does not wait on human approval. Energy, carbon, cost, water, WUE, average PUE, average CPU and memory load, cooling efficiency and the count of underutilized servers are all calculated directly from raw telemetry over a selected date range, whether or not an operator has acted on anything during that window. A separate section of the same report tracks the decision side: how many recommendations were accepted, snoozed or dismissed and what the accepted ones were estimated to save in energy, carbon, cost and storage at the moment each was approved. Reports are downloadable in both CSV and PDF form.

IV. IMPLEMENTATION

The methodology described in Section III was implemented as a working prototype, covering the full pipeline from telemetry ingestion through to the human decision interface.

A. Backend Implementation

The backend is implemented in FastAPI with SQLAlchemy over PostgreSQL, exposing telemetry ingestion, analytics, forecasting, recommendation and ESG reporting endpoints

Each of the four simulated monitors post to its own ingestion endpoint, where they are validated before being written to their respective telemetry table.

TABLE I. DATABASE BACKEND

No	Table	Purpose
1.	Servers	Server registry - static identity / metadata for each of the 13 simulated servers
2.	Server telemetry	Raw compute readings, posted every 15s
3.	Power telemetry	Raw power readings, posted every 15s
4.	Cooling telemetry	Raw cooling readings, posted every 15s
5.	Storage telemetry	Raw storage readings posted every 30s
6.	Forecasts	Persisted CPU forecasts (trained model at one hour, historical average at six and twenty-four hours) with each prediction's algorithm and validation metrics.
7.	Flags	Active / cleared rule based waste detection (idle, stale, duplicate, over provisioned)
8.	Recommendations	Ranked actions generated from a flag, decision-support only
9.	Operator actions	The operator's decision and the what-if impact estimate at that exact moment
10.	Preferences	Single-row, live editable scoring weights used by the recommendation ranking formula
11.	Users	To store Role based access and sign in accordingly.
12.	Site Settings	Single-row, operator-editable config - grid emission factor plus declared values telemetry can't derive (installed capacity, renewable/reuse factors)

B. Frontend Implementation

The frontend is a React single-page application with thirteen routed pages.

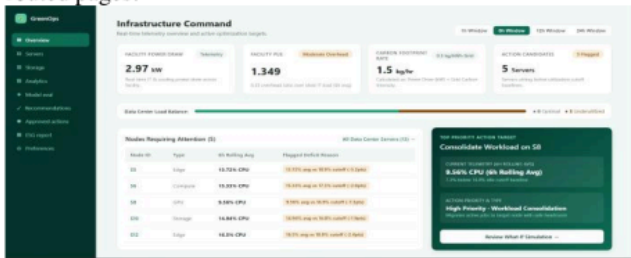


Fig. 6. Dashboard Overview page

The Overview page summarizes facility wide values for PUE, cumulative energy, carbon and the current count of flagged servers being refreshed from live simulated telemetry.

The Servers page shows the full server inventory filterable by type, region and status (Healthy/Underutilized/Stale/No Data), giving an operator a single screen to scan everything before drilling into any one machine. The Server detail page gives insights into per-server view combining live

environmental reading, the CPU forecast and any active waste flags for that specific server.

The Storage page shows specific waste view, surfacing stale, duplicated and over-provisioned volumes flagged by the rule engine.

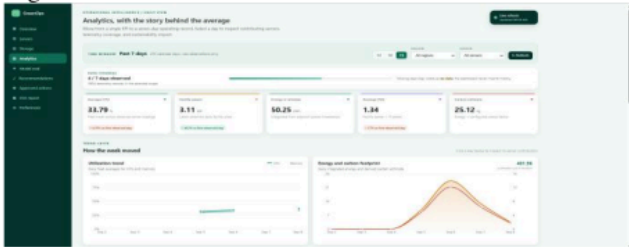


Fig. 7. Analytics page

The Analytics page breaks down the facility-level of idle servers, power/cooling performance and storage waste over time which is used to spot trends.

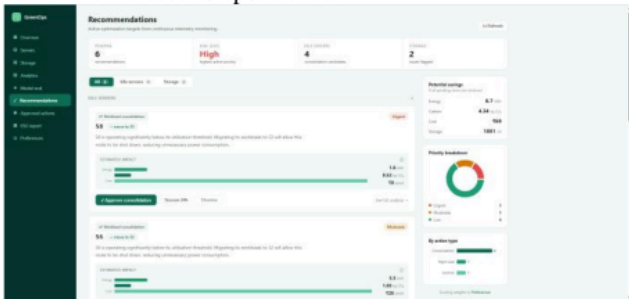


Fig. 8. Recommendations page

The Recommendations page ranks a list of pending consolidate/rightsized actions with inline Accept / Snooze / Dismiss controls, mini impact charts and a sidebar summarizing pending savings across the batch. The Recommendation details page gives full insights about a single recommendation. It ranks all the candidate tables with a CPU history chart, a storage breakdown and the ability to switch candidates and see the impact estimate re-scale live.

The Approved actions logs the past operator's decisions, filterable by executed vs awaiting executions where an operator can retroactively attach an executed at timestamp and a note once the action is carried out.

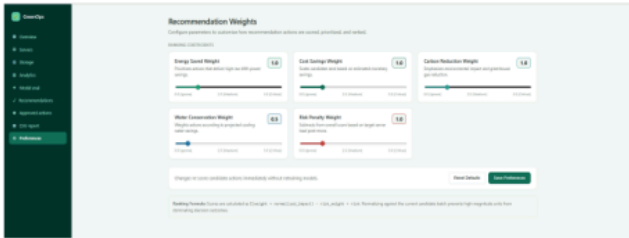


Fig. 9. Preferences page

The Preferences page has a weight slider for cost, carbon, water and risk that feeds directly into the recommendation scoring formula, re-ranking pending recommendations instantly without retraining any model.



Fig. 10. ESG reporting page 1

This page has aggregated log of operator-action over a selected date range, exportable as CSV or PDF, presented explicitly as a decision record.

V. RESULTS AND EVALUATION

A. CPU and Power Estimation

Both estimation models were trained on the Kaggle “Green AI Data Center Telemetry” dataset (10,000 records and 15 columns), split by 80/20 with a most-frequent category imputer handling missing values in data center region (3,990 missing entries). The selection rule kept Linear Regression as the default and only promoted Random Forest or XGBoost if either beat it by more than 5% relative MAE.

TABLE II. CPU ESTIMATION

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	4.511	7.671	0.845
Random Forest	4.828	8.077	0.828
XGBoost	4.833	7.999	0.831

TABLE III. POWER ESTIMATION

Model	MAE (kW)	RMSE (kW)	R^2
Linear Regression	0.211	0.330	0.422
Random Forest	0.227	0.346	0.366
XGBoost	0.222	0.341	0.385

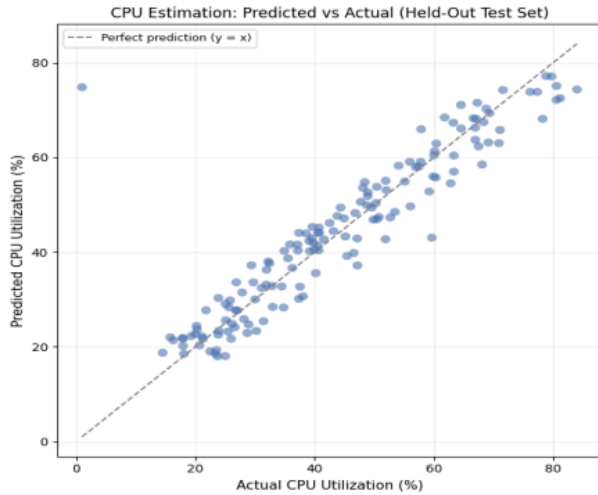


Fig. 11. CPU Estimation: Predicted vs Actual

Fig.11 demonstrates the practical accuracy of our model, where an R^2 value of 0.845 shows a strong alignment between predicted and actual CPU utilization. The data points consistently follow the ideal prediction line, with minor variance isolated to the extreme values where training samples were naturally sparse.

B. Workload Forecasting

The forecasting models were trained separately using telemetry accumulated by the live simulator.

TABLE IV. CPU WORKLOAD FORECAST

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	6.514	8.090	0.886
Random Forest	6.703	8.392	0.877
XGBoost	6.567	8.178	0.883

TABLE V. MEMORY WORKLOAD FORECAST

Model	MAE (%)	RMSE (%)	R^2
Linear Regression	6.560	8.185	0.851
Random Forest	6.843	8.527	0.839
XGBoost	6.795	8.489	0.840

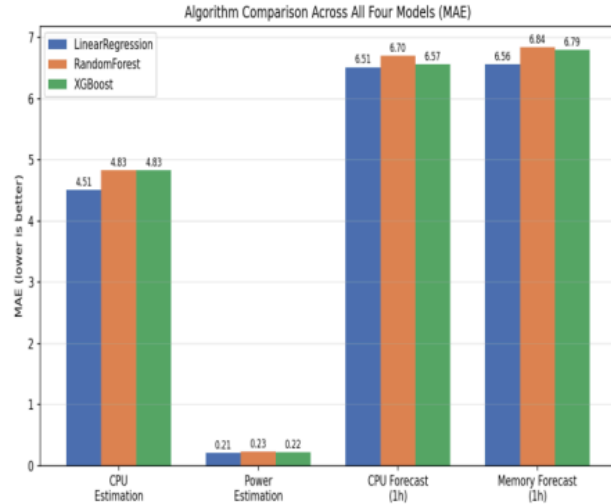


Fig. 12. Comparing algorithms across all four models using MAE

Across all three algorithms the margin Linear Regression won all four targets by the selection rule’s 5% margin. So the simpler model was kept throughout as shown in Fig. 12.

VI. CONCLUSION AND FUTURE WORK

GreenOps demonstrates that the capabilities most of the reviewed literature treats separately with continuous observability, ML-based prediction, rule-based waste detection and sustainability reporting can run as one pipeline with a single human decision point at the end. The working prototype validates this end to end with estimation models trained and honestly evaluated on real telemetry, forecasting models trained

on the system's own accumulated history, a four-dimension what-if safety check that screens every migration option before it's ever surfaced and role-based review that ties each decision to a specific accountable person rather than to automation.

Future work will focus on replacing the simulated telemetry with real integrations (Prometheus, smart PDU's/BMS, API's) and swapping the placeholder carbon-intensity and WUE constants for live regional grid and facility water data. The what-if check should also gain workload criticality gating, along with SLA requirements and storage and electrical capacity limits, so that no recommendation is surfaced for a workload that is too critical to move. Longer-range planning needs a 24-hour forecast. That's not possible yet, since there isn't enough telemetry to train it on. Nothing currently checks whether an approved migration saved what was predicted either, so a before-and-after comparison of power readings is worth adding. Beyond that, GreenOps could be tried across several data centers and watched over a longer period, to see whether operators come to trust it as its track record grows.

VII. REFERENCES

- [1] Council on Energy, Environment and Water (CEEW), "How is data centre infrastructure in India shaping power and water use?," 2026.
- [2] A. Beloglazov, J. Abawajy and R. Buyya, "Energy-aware resource allocation heuristics for efficient management of data centers for cloud computing," *Future Generation Computer Systems*, vol. 28, no. 5, pp. 755–768, 2012.
- [3] C. Subramanian, A. Vasan and A. Sivasubramaniam, "Reducing data center power with server consolidation: Approximation and evaluation," in *Proc. 17th Int. Conf. High Performance Computing (HiPC)*, IEEE Computer Society, 2010.
- [4] G. Lovász, F. Niedermeier and H. de Meer, "Performance tradeoffs of energy-aware virtual machine consolidation," *Cluster Computing*, vol. 16, no. 3, pp. 481–496, 2013.
- [5] M. Uddin *et al.*, "Server consolidation: A technique to enhance cloud data center power efficiency and overall cost of ownership," *International Journal of Distributed Sensor Networks*, 2021.
- [6] J. Bawa, K. K. Chahal and K. Kaur, "Improving cloud resource management: An ensemble learning approach for workload prediction," *The Journal of Supercomputing*, vol. 81, no. 10, 2025.
- [7] M. R. R. Deva and N. Jain, "Utilizing Azure Automated Machine Learning and XGBoost for predicting cloud resource utilization in enterprise environments," in *Proc. 2025 Int. Conf. Networks and Cryptology (NETCRYPT)*, IEEE, 2025.
- [8] T. Khan, W. Tian, S. Ilager and R. Buyya, "Workload forecasting and energy state estimation in cloud data centres: ML-centric approach," *Future Generation Computer Systems*, vol. 128, pp. 320–332, 2022.
- [9] A. Radovanović *et al.*, "Carbon-aware computing for datacenters," arXiv preprint arXiv:2106.11750, 2021.
- [10] X. Zhan *et al.*, "Data center cooling system optimization using offline reinforcement learning," arXiv preprint arXiv:2501.15085, 2025.
- [11] M. Alex, S. O. Ojo and F. M. Awuor, "Carbon-aware, energy-efficient and SLA-compliant virtual machine placement in cloud data centers using deep Q-networks and agglomerative clustering," *Computers*, vol. 14, no. 7, Art. no. 280, 2025.
- [12] M. Grochulska-Salak *et al.*, "Green data centres: Sustainable solutions with green energy and green-blue infrastructure," *Energies*, vol. 18, no. 24, Art. no. 6592, 2025.
- [13] European Commission, "Corporate sustainability reporting," Financial Markets, Company Reporting and Auditing.
- [14] Council on Energy, Environment and Water (CEEW) and SYSTEMIQ, "Scaling India's data centre ecosystem," 2026.
- [15] Y. Tao and P. Gao, "Global data center expansion and human health: A call for empirical research," *Eco-Environment & Health*, vol. 4, no. 3, Art. no. 100157, 2025.
- [16] J. Gorey, "Data drain: The land and water impacts of the AI boom," *Land Lines Magazine*, Lincoln Institute of Land Policy, Oct. 17, 2025.
- [17] V. Vaccaro, L. C. Tagliabue and M. Aldinucci, "Sustainable data centers: Advancing energy efficiency and resource optimization," in *Proc. 33rd Euromicro Int. Conf. Parallel, Distributed and Network-Based Processing (PDP)*, Turin, Italy, 2025, pp. 486–493.
- [18] A. Ahmed, D. Brown and A. Gegov, "Dynamic resource allocation through workload prediction for energy efficient computing," in *Advances in Computational Intelligence Systems*, P. Angelov, A. Gegov, C. Jayne and Q. Shen, Eds., Springer, 2017, pp. 35–44.

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